```
client 2 writes, Nagle on          p50   41.00  p99   41.10 ms
client 2 writes, TCP_NODELAY       p50    0.07  p99    0.12 ms
```

A companion to the Testing Field Guide

Java *performance* deep dives

TCP_NODELAY and delayed ACKs, connections and timeouts, pools and Little's Law, measuring latency, the JIT, GC pauses, virtual threads, containers, the memory model and Kafka — each explained from the specification or the source code, and each measured. Every number in this book is read from a log written in CI.

11	26	42	1
chapters	experiments	logs	runner, named

INSIDE

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Facts as of October 2026, for JDK 21 unless a page says otherwise. Every entry, page reference and footer link is clickable.

COLOUR LANGUAGE

Network · queues and measurement · the JVM · the platform

READ WITH

Testing Java applications — Field Guide (Testing-Field-Guide.pdf, same repository). Its performance chapter introduces several of these topics in a page each; this book is the long version.

ON EVERY PAGE

The mechanism, from the specification or the source. The experiment's code, as it ran. The real output. What to do about it. The sources, at the foot.

BEFORE YOU START

How it was measured

Each page explains a mechanism and then measures it. Explanations cite a specification, the source code of the version discussed, or the vendor's documentation; measurements come from programs in this repository, run in CI.

Sources for explanations

The specification (RFC 9293 for TCP, the Java Language Specification for the memory model, JEPs for JDK features), the **source code of the version discussed** — OpenJDK 21, Linux 6.17, the libraries — or the vendor's documentation. Each page lists what it used at its foot; the full list is at the back.

Measurements

Each chapter has a small program in `deepdive/`. The `deepdive` CI job runs all of them on every push (`deepdive/run.sh`) and publishes the logs to the `ci-logs-deepdive` branch. This book is built from those logs: every measured number on a page is read from them at build time.

The machine

Runner	GitHub Actions, `ubuntu-latest`
CPU	4 CPUs · AMD EPYC 7763 64-Core Processor
Kernel	Linux 6.17.0-1022-azure, `CONFIG_HZ=1000`
JDK	OpenJDK 21.0.12.1 (Temurin)
`ip_local_port_range`	32768 60999
`tcp_tw_reuse`	2
`tcp_syn_retries`	6
`tcp_pingpong_thresh`	1
`tcp_fin_timeout`	60

From `machine.txt`, written by the same run.

The experiments

CH.	PROGRAM	LOG
1	tcp/Nagle.java + tcpdump	nagle.txt, nagle-trace.txt
2	tcp/Connections.java	connections.txt
3	tcp/Timeouts.java, SynRetries.java + iptables	timeouts.txt, syn*.txt
4	pool/PoolSize.java, CpuPool.java	pool.txt, cpupool.txt
5	latency/CoordinatedOmission.java	co.txt
6	jit/WarmUp.java, DeadCodeBench.java (JMH)	warmup*.txt, jmh.txt, jit-flags.txt
7	gc/GcLatency.java, G1 and ZGC	gc-*.txt, gc-*.log
8	vt/VirtualThreads.java	vt.txt
9	box/Resources.java, Throttle.java in Docker	containers.txt, throttle.txt
10	jmm/StaleFlag.java + 4 jcstress tests	staleflag.txt, jcstress.log
11	kafka/KafkaLatency.java, one broker in Docker	kafka-*.txt

One number, traced

STEP	EVIDENCE
The claim (chapter 1)	Two small writes with Nagle on: median **41.00 ms** per request
The log line	`client 2 writes, Nagle on p50 41.00` — `nagle.txt`, written by `Nagle.java` in CI
The packets	`nagle-trace.txt`: the server's ACK of the first 50-byte half left 40.88 ms after it; the second half was held until then — tcpdump on loopback
Why 40 ms	Linux `include/net/tcp.h`: `TCP_DELACK_MIN = HZ/25`; with `CONFIG_HZ=1000` that is 40 ms
Why it waits at all	RFC 9293 §3.7.4 (Nagle: hold small segments while data is unacknowledged) and §3.8.6.3 (delayed ACK)

- **The text follows the data.** Where a sentence interprets a result — "slower", "removed by the JIT", "not caused by a GC pause" — it is computed from the log, and changes if the result changes.
- **Conclusions are checked.** `guide/claims_deep.py` re-reads the logs and fails the build if a conclusion this book states no longer holds.
- **One machine.** The numbers are from one CI runner and depend on its CPU, kernel and JDK. To rerun on another machine: `bash deepdive/run.sh` (Linux, JDK 21, Maven, Docker; `sudo` for tcpdump and iptables).
- **Versions matter.** Statements are for JDK 21 and the library versions named; where a later JDK changed the behaviour (JEP 474, 490, 491) the page says so.

01 · NAGLE AND DELAYED ACK

Nagle's algorithm and delayed ACK

TCP has two mechanisms that reduce the number of small packets. The **sender** holds back a small segment while earlier data is unacknowledged (Nagle). The **receiver** holds back its ACK, hoping to piggyback it on a reply (delayed ACK). In a request/response exchange where one message goes out in two `write()` calls, each waits for the other — until a timer fires.

Nagle's algorithm — the sender

> "If there is unacknowledged data (i.e., SND.NXT > SND.UNA), then the sending TCP endpoint buffers all user data (regardless of the PSH bit) until the outstanding data has been acknowledged or until the TCP endpoint can send a full-sized segment." — RFC 9293 §3.7.4

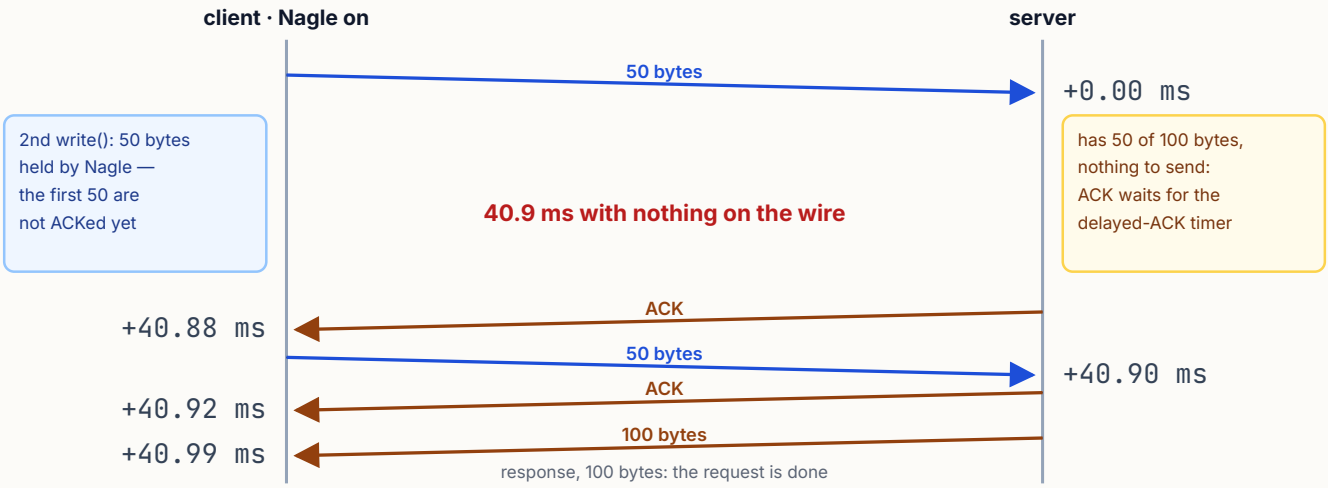
TCP **SHOULD** implement it (SHLD-7); there **MUST** be a way to disable it per connection (MUST-17). In Java that is `TCP_NODELAY`, off by default on a `java.net.Socket`.

Delayed ACK — the receiver

> "A TCP endpoint SHOULD implement a delayed ACK (SHLD-18) ... the delay MUST be less than 0.5 seconds (MUST-40)." An ACK SHOULD go out for at least every second full-sized segment. — RFC 9293 §3.8.6.3

Linux starts each connection in **quick-ACK** mode and leaves it once the connection looks like request/response ("ping-pong"): a reply sent within the ACK timeout of receiving data counts. From then on, a lone segment waits for the timer.

The stall, captured



LINUX	VALUE	MEANING	WHERE
`TCP_DELACK_MIN`	HZ/25 jiffies = **40 ms**	shortest delayed-ACK timer	include/net/tcp.h
`TCP_DELACK_MAX`	HZ/5 jiffies = **200 ms**	longest delayed-ACK timer	include/net/tcp.h
`TCP_MAX_QUICKACKS`	**16**	ACKs sent at once in quick-ACK mode	include/net/tcp.h
`tcp_pingpong_thresh`	**1** on the runner	replies before a connection counts as request/response	ip-sysctl.rst
`CONFIG_HZ`	**1000** on the runner	timer tick: 1 ms	/boot/config

The capture shows the second request on the connection. The server's ACK left **40.9 ms** after the first half of the request; `TCP_DELACK_MIN` is 40 ms on this kernel.

The **first** request did not wait: its ACK came back in 0.015 ms, while the new connection was still in quick-ACK mode. After one request and its reply, Linux treats the connection as ping-pong and starts delaying ACKs.

SOURCES RFC 9293 (2022) §3.7.4, §3.8.6.3 · RFC 896 (Nagle, 1984) · Linux include/net/tcp.h, net/ipv4/tcp_input.c, tcp_output.c (tcp_event_data_sent: pingpong) · Documentation/networking/ip-sysctl.rst · OpenJDK 21 StandardSocketOptions.TCP_NODELAY · capture: tcpdump on loopback, GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1

01 · NAGLE AND DELAYED ACK

Measured: one or two writes per message, 7 setups

One connection on loopback, 200 request/response round trips of 100 bytes each way. The only differences: whether a message is written in one `write()` or two (like headers, then body), on which side, and which socket option is set.

Nagle.java — the client side

```java
static long[] exchange(Socket socket, int writes, int rounds)
    throws Exception {
  OutputStream out = socket.getOutputStream();
  InputStream in = socket.getInputStream();
  long[] rtt = new long[rounds];
  for (int i = 0; i < rounds; i++) {
    long t0 = System.nanoTime();
    write(out, writes);
    in.readNBytes(100);
    rtt[i] = System.nanoTime() - t0;
  }
  return rtt;
}

/** 100 bytes in one write, or 50 + 50 like headers + body. */
static void write(OutputStream out, int writes)
    throws Exception {
  if (writes == 1) {
    out.write(new byte[100]);
  } else {
    out.write(new byte[50]);
    out.write(new byte[50]);
  }
  out.flush();
}
```

SETUP	P50 MS	P99 MS	MAX MS	OVER 30 MS
client 2 writes, Nagle on	41.00	41.10	41.19	199 of 200
client 2 writes, TCP_NODELAY	0.07	0.12	0.15	0 of 200
client 1 write, Nagle on	0.05	0.08	0.23	0 of 200
server 2 writes, Nagle on	41.00	41.15	41.98	199 of 200
server 2 writes, TCP_NODELAY	0.05	0.09	0.18	0 of 200
client 2 writes, QUICKACK once	41.00	41.08	41.18	199 of 200
client 2 writes, QUICKACK per read	0.06	0.11	0.15	0 of 200

Real output · GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1

TCP_NODELAY defaults in common libraries

LIBRARY	TCP_NODELAY	EVIDENCE IN ITS SOURCE
JDK `HttpClient`	on	`PlainHttpConnection`: `setOption(TCP_NODELAY, true)`
JDK `HttpServer`	**off**	only with `-Dsun.net.httpserver.nodelay=true` (`ServerConfig`)
`java.net.Socket`	**off**	`StandardSocketOptions.TCP_NODELAY`: default `FALSE`
Tomcat	on	`SocketProperties.tcpNoDelay = TRUE`
Jetty 12	on	`ServerConnector._acceptedTcpNoDelay = true`
Netty 4.1	on	`DefaultSocketChannelConfig`: on unless Android
PostgreSQL JDBC	on	`PGProperty.TCP_NO_DELAY`: default `"true"`
Lettuce (Redis)	on	`SocketOptions.DEFAULT_SO_NO_DELAY = true`
Kafka client	on	`Selector`: `socket.setTcpNoDelay(true)`

- **Two writes per message + Nagle = one timer per message.** Every round trip took about 41 ms; every fix brought p99 under 0.1 ms.
- **Write each message once.** Build it in a buffer (or wrap the stream in a `BufferedOutputStream`) and flush once: Nagle never has a second small segment to hold.
- **Or set `TCP_NODELAY`** on request/response connections — on the side that writes twice. Most servers and the JDK client already do; the JDK's own `HttpServer` does not.
- **The receiver can help:** `jdk.net.ExtendedSocketOptions.TCP_QUICKACK` (Linux, JDK 10+) asks for an immediate ACK — but it is not sticky. Set once, nothing changed; set before every read, the stall was gone.
- **Symptom:** a p50 just above 40 ms on Linux while the CPUs are idle. The main guide's load test of the JDK `HttpServer` showed the same stall.

SOURCES OpenJDK 21: jdk/internal/net/http/PlainHttpConnection.java, sun/net/httpserver/ServerConfig.java, ServerImpl.java, java/net/StandardSocketOptions.java, jdk/net/ExtendedSocketOptions.java (TCP_QUICKACK, since 10) · Tomcat main: SocketProperties.java · Jetty 12.0: ServerConnector.java · Netty 4.1: DefaultSocketChannelConfig.java · pgjdbc: PGProperty.java · Lettuce: SocketOptions.java · Kafka: clients/.../network/Selector.java · results: deepdive/logs/nagle.txt

01 · NAGLE AND DELAYED ACK

When Nagle helps, and what each setting changes

Nagle's algorithm trades a delay for fewer, fuller segments. Chapter 1 measured its cost in a request/response exchange; this page measures a one-way stream of small writes and lists the settings that change the behaviour.

Linux 6.17 net/ipv4/tcp_output.c — the comment above tcp_nagle_check()

```
/* Return false, if packet can be sent now without violation Nagle's rules:
 * 1. It is full sized. (provided by caller in %partial bool)
 * 2. Or it contains FIN. (already checked by caller)
 * 3. Or TCP_CORK is not set, and TCP_NODELAY is set.
 * 4. Or TCP_CORK is not set, and all sent packets are ACKed.
 *    With Minshall's modification: all sent small packets are ACKed.
 */
```

"Full sized" means one maximum segment size (MSS). Rule 4 with Minshall's modification: a small segment may go out once every earlier *small* segment is acknowledged, even while full-sized ones are in flight (`tcp_minshall_check`).

Measured: 10,000 one-byte writes, no reply

WRITER	TCP_AUTOCORKING	MS	DATA SEGMENTS SENT
Nagle on (default)	1	44	2,183
TCP_NODELAY	1	52	5,553
BufferedOutputStream, one flush	1	3	2
Nagle on (default)	0	45	2,190
TCP_NODELAY	0	88	9,606
BufferedOutputStream, one flush	0	2	2

SmallWrites.java · the client socket's `data_segs_out` counter (`ss -ti`) · GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1

- **Nagle on**: 2,183 data segments for 10,000 writes — bytes waited while earlier ones were unacknowledged, then went together.
- **TCP_NODELAY, autocorking off**: 9,606 data segments.
- **TCP_NODELAY, autocorking on** (the default): 5,553 — `tcp_autocorking` merged writes while earlier packets were still queued.
- **One buffered write**: 2 segments in 3 ms.

The settings

SETTING	WHERE IT IS SET; FROM JAVA	EFFECT
`TCP_NODELAY`	socket; `Socket.setTcpNoDelay`, `StandardSocketOptions.TCP_NODELAY` (default false)	rule 3: a small segment may go out while others are unacknowledged
`TCP_QUICKACK`	socket, Linux; `jdk.net.ExtendedSocketOptions.TCP_QUICKACK`	the receiver ACKs at once; "this option is not permanent" (its Javadoc) — chapter 1 set it before every read
`TCP_CORK`	socket, Linux; not in JDK 21's `ExtendedSocketOptions`	holds partial segments until uncorked (rule 3 and 4 require it unset)
`tcp_autocorking`	sysctl, default 1	coalesces consecutive small writes "if at least one prior packet for the flow is waiting in Qdisc queues or device transmit queue"

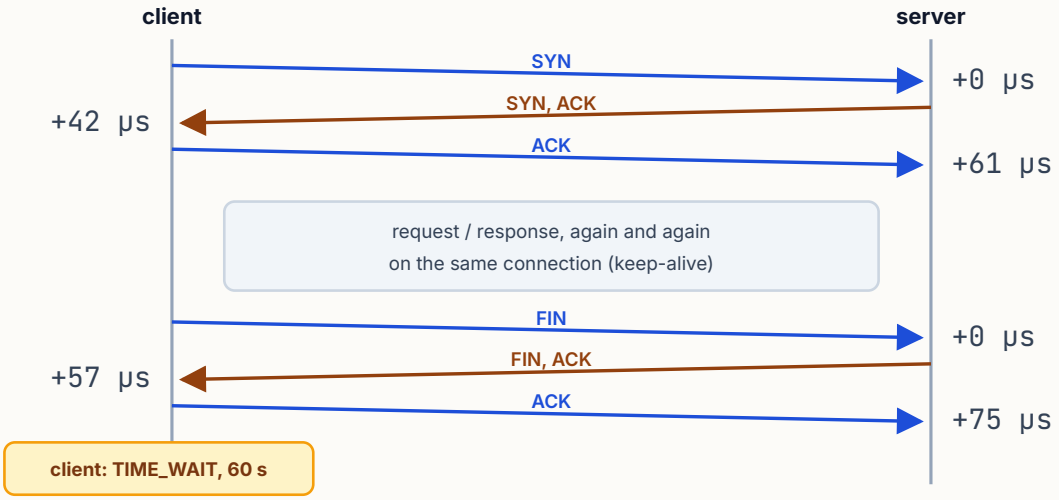
- Set `TCP_NODELAY` when a request or response is written in more than one `write()` and the peer waits for the whole message — chapter 1's case: 41 ms per exchange with Nagle on, under 0.2 ms with it off.
- **Leave Nagle on** for a one-way stream of small writes where no reply is awaited: here it cut the segment count.
- **Prefer writing each message once** (build it in a buffer): then neither Nagle nor `TCP_NODELAY` matters, and the segment count is lowest.

SOURCES Linux 6.17 net/ipv4/tcp_output.c (tcp_nagle_check, tcp_minshall_check), include/net/tcp.h (TCP_NAGLE_OFF/CORK/PUSH) · Documentation/networking/ip-sysctl.rst (tcp_autocorking) · OpenJDK 21 jdk/net/ExtendedSocketOptions.java, java/net/StandardSocketOptions.java · results: deepdive/logs/smallwrites.txt

02 · CONNECTIONS

Connection set-up, reuse and close

Before the first byte of a request, TCP spends a round trip on the handshake. After the last byte, the side that closes **first** keeps the connection in TIME_WAIT, holding its port. Both costs are paid once per connection, however many requests it carries.



TIME_WAIT

> "When a connection is closed actively, it MUST linger in the TIME-WAIT state for a time 2xMSL (Maximum Segment Lifetime) (MUST-13)." "For this specification the MSL is taken to be 2 minutes." — RFC 9293 §3.6.1, §3.4.2

Sequence numbers stay "busy" until MSL seconds have passed (§3.4.3): the wait keeps old segments from being taken for a new connection's. Linux does not wait 4 minutes but a fixed **60 s** (`TCP_TIMEWAIT_LEN`). `tcp_fin_timeout` is something else — how long an orphaned connection may sit in FIN_WAIT_2.

Ephemeral ports run out

A connection is identified by four values: source address and port, destination address and port. To one destination, a client can only vary its own port. The runner's range is `32768-60999` : **28,232** ports. Each closed connection keeps its port for 60 s, so one client address can open at most about **471 new connections per second** to one server address and port, for as long as that lasts — unless TIME_WAIT reuse applies.

Reuse is not on everywhere

`tcp_tw_reuse` lets a new outgoing connection take a port still in TIME_WAIT when that is safe. Its default is **2: loopback traffic only** — the runner has `2` . With the default, a client connecting to `localhost` can reuse ports in TIME_WAIT; a client connecting to another host cannot.

TCP states shown by `ss`

STATE	MEANS	A LARGE COUNT INDICATES
`SYN-SENT`	connect() sent a SYN, no answer yet	the server is unreachable or dropping packets — see the next chapter
`ESTABLISHED`	open, data can flow both ways	the number of open connections; compare with pool sizes
`FIN-WAIT-2`	we closed; the peer has not closed its side yet	the peer is not closing its sockets
`CLOSE-WAIT`	the peer closed; our application has not called close()	**a socket leak in our code**
`TIME-WAIT`	we closed first; waiting before the port can be reused	many short-lived connections closed by this side

SOURCES RFC 9293 §3.3.2, §3.4.2, §3.4.3, §3.6.1 · Linux include/net/tcp.h (TCP_TIMEWAIT_LEN) · Documentation/networking/ip-sysctl.rst (ip_local_port_range, tcp_tw_reuse, tcp_fin_timeout) · capture and sysctl values: GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1

02 · CONNECTIONS

Measured: kept-alive and per-request connections

2,000 request/response exchanges on loopback: over one kept-alive connection, then over a new connection each time, closed four different ways. Afterwards the program counts sockets by state in the kernel's own tables, `/proc/net/tcp` and `tcp6`.

HOW	MS / REQUEST	VS KEPT-ALIVE	TIME_WAIT CLIENT / SERVER	CLOSE_WAIT CLIENT / SERVER	FIN_WAIT2 CLIENT / SERVER
one kept-alive connection	59.3	1.0×	–	–	–
new connection, client closes first	266.4	4.5×	**2000** / 0	0 / 0	0 / 0
new connection, server closes first	252.3	4.3×	0 / **2000**	0 / 0	0 / 0
new connection, client closes with `SO_LINGER(0)`	246.2	4.2×	0 / 0	0 / 0	0 / 0
new connection, server never closes (a leak)	232.2	3.9×	0 / 0	0 / **2000**	**2000** / 0

Real output · GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1

Connections.java — a new connection per request

```
static void perRequest(Close how) throws Exception {
  try (var server = new ServerSocket(0, 512, LO)) {
    int port = server.getLocalPort();
    Thread.ofPlatform().daemon()
        .start(() -> oneEach(server, how));
    long t0 = System.nanoTime();
    for (int i = 0; i < N; i++) {
      try (var s = new Socket(LO, port)) {
        s.setTcpNoDelay(true);
        if (how == Close.CLIENT_RESET) {
          s.setSoLinger(true, 0);       // close() sends RST
        }
        s.getOutputStream().write(new byte[100]);
        var in = s.getInputStream();
        in.readNBytes(100);
        if (how == Close.SERVER_FIRST) {
          in.read();          // wait for the server's FIN
        }
      }
    }
    Thread.sleep(200);
    report("new connection, " + how.name().toLowerCase()
        .replace('_', ' '), t0, port);
  }
}
```

- **A new connection per request cost 4.5× as much as a kept-alive one**, on loopback. Across a network each new connection also adds the handshake's round trip.
- **TIME_WAIT was on the side that closed first**: 2,000 sockets on that side, none on the other.
- **`SO_LINGER(0)` left no TIME_WAIT.** With a zero linger time, Linux `tcp_close()` calls `tcp_disconnect()`, which sends a reset (RST) instead of a FIN; unsent data is discarded.
- **A server that never closed** left 2,000 sockets in CLOSE_WAIT, and the clients' ends in FIN_WAIT2 — the state `tcp_fin_timeout` limits for orphaned sockets. CLOSE_WAIT means the application has not called `close()`.
- **The JDK `HttpClient` keeps idle connections for 30 s** (`jdk.httpclient.keepalive.timeout`) and reuses them; a new client per request opens a new connection per request.

- **Count sockets by state** during a test with `ss -tan state time-wait` (or `close-wait`), or from `/proc/net/tcp` as the program on the right does.

counting sockets by state

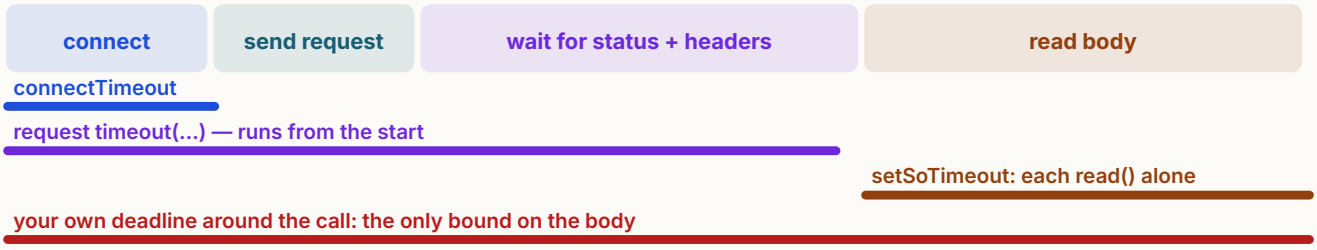
```
static long count(String state, int port, boolean local)
    throws Exception {
  String hex = String.format(":%04X", port);
  long n = 0;
  for (String table : new String[] {"tcp", "tcp6"}) {
    Path p = Path.of("/proc/net/" + table);
    if (!Files.exists(p)) {
      continue;                  // no IPv6 on this machine
    }
    n += Files.readAllLines(p).stream()
        .skip(1).map(l -> l.trim().split("\\s+"))
        .filter(f -> f[3].equals(state))
        .filter(f -> (local ? f[1] : f[2]).endsWith(hex))
        .count();
  }
  return n;
}
```

SOURCES OpenJDK 21 java.net.http module-info.java (jdk.httpclient.keepalive.timeout, default 30) · Linux ip-sysctl.rst · results: deepdive/logs/connections.txt

03 · TIMEOUTS

What each timeout covers

An HTTP call has phases: connect, send, wait for the status line and headers, read the body. Each timeout setting bounds one phase; a phase without one waits indefinitely. The table gives the bound for each phase from the JDK 21 Javadoc and source; the next page measures them.



PHASE	BOUNDED BY	IF UNSET
TCP connect (handshake)	`HttpClient.Builder.connectTimeout` ; `Socket.connect(addr, ms)`	the kernel's SYN retries
request sent, waiting for the status line and headers	`HttpRequest.Builder.timeout` (counts from the start, connect included)	forever
reading the body	nothing in `HttpClient` ; `Socket.setSoTimeout` bounds each `read()` , not the total	forever
the whole call	only what you add around it — e.g. `future.get(timeout)`	forever

What the request timeout covers

> "If the response is not received within the specified timeout then an HttpTimeoutException is thrown … The effect of not setting a timeout is the same as setting an infinite Duration, i.e. block forever." — `HttpRequest.Builder.timeout`

"The response" means the status line and headers: in JDK 21 the timer is cancelled when `exch.responseAsync()` completes (`MultiExchange` , step "5. handle errors and cancel any timer set"). A body that arrives slowly is not covered.

Defaults: what you get when you set nothing

CLIENT	CONNECT	WAITING FOR DATA	SOURCE
JDK `HttpClient`	none	none ("block forever")	`HttpClient.Builder` , `HttpRequest.Builder` Javadoc
`HttpURLConnection`	0 = infinite	0 = infinite	`URLConnection` : "A timeout of zero is interpreted as an infinite timeout"
`java.net.Socket`	kernel's SYN retries	0 = infinite, per `read()`	`Socket.setSoTimeout` Javadoc
PostgreSQL JDBC	`connectTimeout` 10 s	`socketTimeout` 0 = none	`PGProperty`
HikariCP	`connectionTimeout` 30 s — waiting for a pool connection	—	`HikariConfig`
Lettuce (Redis)	10 s	command timeout 60 s	`SocketOptions` , `RedisURI.DEFAULT_TIMEOUT`
Kafka producer	—	`request.timeout.ms` 30 s	`ProducerConfig`

SOURCES OpenJDK 21: java/net/http/HttpRequest.java, HttpClient.java (Javadoc) · jdk/internal/net/http/MultiExchange.java, ResponseTimerEvent.java · java/net/Socket.java, URLConnection.java (Javadoc) · pgjdbc PGProperty.java · HikariCP HikariConfig.java · Lettuce SocketOptions.java, RedisURI.java · Kafka ProducerConfig.java

03 · TIMEOUTS

Measured: three unresponsive servers

Three local peers misbehave in different phases. The harness gives up after 5 s — the overall deadline the HTTP client does not provide. Packets to `192.0.2.1` (an address reserved for documentation, RFC 5737) are dropped with `iptables`, so the handshake never completes.

Timeouts.java — one call, two optional timeouts

```
static void call(String name, String hostPort,
    Duration connect, Duration request) throws Exception {
  var client = HttpClient.newBuilder();
  if (connect ≠ null) {
    client.connectTimeout(connect);
  }
  var req = HttpRequest.newBuilder(
      URI.create("http://" + hostPort + "/"));
  if (request ≠ null) {
    req.timeout(request);
  }
  long t0 = System.nanoTime();
  String result;
  try {
    var r = client.build().sendAsync(req.build(),
        BodyHandlers.ofString())
        .get(GIVE_UP_S, TimeUnit.SECONDS);
    result = "returned " + r.statusCode() + ", body "
        + r.body().length() + " bytes";
  } catch (TimeoutException e) {
    result = "still waiting after " + GIVE_UP_S + " s";
  } catch (Exception e) {
    result = e.getCause().getClass().getSimpleName();
  }
  print(name, t0, result);
}
```

PEER	SETTING	WAITED	RESULT
silent: never answers	no timeouts	5.3 s	still waiting after 5 s
silent: never answers	connectTimeout 1 s	5.0 s	still waiting after 5 s
silent: never answers	request timeout 2 s	2.0 s	`HttpTimeoutException`
trickle: headers at once, body 1 byte / 0.5 s	request timeout 2 s	5.0 s	still waiting after 5 s
dropped: packets dropped (no handshake)	no timeouts	5.0 s	still waiting after 5 s
dropped: packets dropped (no handshake)	request timeout 2 s	2.0 s	`HttpConnectTimeoutException`
dropped: packets dropped (no handshake)	connectTimeout 1 s	1.0 s	`HttpConnectTimeoutException`
trickle: headers at once, body 1 byte / 0.5 s	Socket soTimeout 1 s	6.0 s	returned 70 bytes, no error

Real output · GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1

No timeout at all: the kernel decides

real output · connect() with no timeout

```
java.net.ConnectException: Connection timed out after 136.4 s
```

The kernel's SYN retransmission count, read from `ss` every second: 0 at 1 s · 2 at 3 s · 3 at 4 s · 4 at 5 s · 5 at 6 s · 6 at 8 s · 7 at 12 s · 0 at 18 s · 1 at 19 s · 2 at 20 s · 3 at 21 s · 4 at 22 s · 0 at 23 s · 1 at 24 s · 0 at 25 s · 6 at 26 s · 7 at 29 s · 8 at 32 s · 9 at 36 s · 10 at 69 s. With `tcp_syn_retries = 6` the kernel gave up after **136 s**.

Connect timeouts

> "If the connection cannot be established within the given duration, then HttpClient::send throws an HttpConnectTimeoutException … If a new connection does not need to be established … this timeout duration has no effect."
> — `HttpClient.Builder.connectTimeout`

The request timer fires during connect too: `ResponseTimerEvent` throws `HttpConnectTimeoutException` when the connection is not yet connected, else `HttpTimeoutException`.

- **The trickling body beat the 2 s request timeout**: still waiting at 5 s. A raw socket with a 1 s read timeout read it all in 6 s without an error — every byte came within 1 s.
- **A connect timeout alone does not help** against a server that accepts and stays silent: still waiting at 5 s.
- **Set both, and a deadline:** `connectTimeout` for the handshake, `timeout` for the headers, and a bound on the whole call around it (`future.get(…)`) — in this run, the only bound that ended the trickling call.

SOURCES RFC 5737 (documentation addresses) · Linux ip-sysctl.rst (tcp_syn_retries) · results: deepdive/logs/timeouts.txt, syn.txt, syn-ss.txt

04 · POOLS AND LITTLE'S LAW

Little's Law and pool size

Every pool is a queue in front of a resource. Little's Law ties the three numbers you can measure together, for any stable system, whatever the pattern of arrivals: **L = λ·W** — the average number of requests inside equals the rate they leave times the average time each spends inside.

Three consequences

1. A pool caps throughput: with C connections each busy for S seconds, at most C/S requests per second leave. Our database (4 slots, ~10.2 ms per query as measured) tops out near 394 per second.

2. Past the cap, more callers only add waiting: 64 callers in a system that serves 394/s means W = 64 / 394 ≈ 162 ms each, however the queue is arranged.

3. A bigger pool moves the queue from the pool into the database: measured on the next page as time spent waiting inside the database.

How big, then?

> "connections = ((core_count * 2) + effective_spindle_count)" — a starting point from the PostgreSQL project, quoted by HikariCP's "About Pool Sizing"

It is a starting value. The next page measures throughput and latency as the pool size changes, for a pool in front of a database and for a pool of CPU-bound threads.

The experiment



PoolSize.java — one caller's loop

```
callers[i] = Thread.ofVirtual().start(() → {
  while (System.nanoTime() < end) {
    long t0 = System.nanoTime();
    inSystem.incrementAndGet();
    pool.acquireUninterruptibly();
    long t1 = System.nanoTime();
    db.acquireUninterruptibly();   // the database's queue
    dbWait.add(System.nanoTime() - t1);
    sleep(QUERY_MS);
    db.release();
    pool.release();
    inSystem.decrementAndGet();
    long t2 = System.nanoTime();
    done.add(new Done(t2 - start, t2 - t0));
  }
});
```

Using it: three questions, answered from the measured numbers

QUESTION	LITTLE'S LAW	WITH THIS CHAPTER'S DATABASE
How many connections are busy at a target rate?	L = λ · S (S = time a query holds a connection)	197 queries/s × 10.2 ms = **2.0** connections busy on average — a pool of 4 has headroom
What would a smaller pool deliver?	λ_{max} = C / S	C = 1: **99/s**, C = 2: **197/s** predicted; measured 98/s and 197/s
How long will callers wait once it is saturated?	W = L / λ_{max}	64 callers / 394/s = **162 ms**; measured mean with pool 4: 162.6 ms

S is derived from the saturated rows only (4 slots ÷ 394 queries/s); the smaller pools are then a prediction, checked against their own rows. The law needs averages over a stable period; it says nothing about the tail — p99 needs the distribution (chapter 5).

SOURCES J. D. C. Little, "A Proof for the Queuing Formula: L = λW", Operations Research 9(3), 1961 · HikariCP wiki, "About Pool Sizing" · deepdive/src/.../pool/PoolSize.java

04 · POOLS AND LITTLE'S LAW

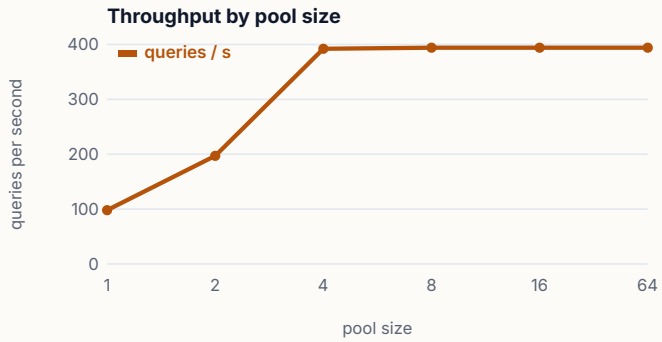
Measured: database pool and CPU pool sizes

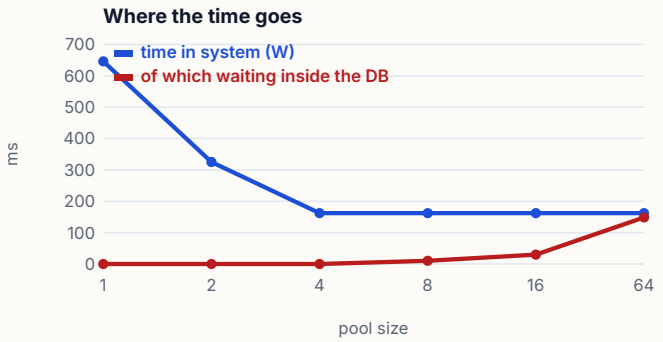
Each row ran for 3 s with 64 callers. λ, W and L were measured independently over the steady window 0.5–2.5 s: λ from completions, W from each request's time in the system, L by counting callers inside every millisecond.

POOL	QUERIES/S (λ)	MEAN MS (W)	P99 MS	WAIT IN DB	L	λ·W
1	98	645.9	654.2	0.0 ms	64.0	63.3
2	197	325.0	325.8	0.0 ms	64.0	64.0
4	392	162.6	162.9	0.0 ms	64.0	63.7
8	394	162.1	162.4	10.1 ms	64.0	63.9
16	394	162.2	162.5	30.2 ms	64.0	63.9
64	394	162.3	162.6	148.4 ms	64.0	64.0

Real output · GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1

- **Little's Law held in every row:** λ·W was within 1.1% of the measured L.
- **Throughput stopped at the database:** from pool size 4 up, 392–394 queries/s.
- **Bigger pools moved the queue into the database:** the wait inside it grew from 0 to 148 ms per query while the total time stayed the same.
- **Too small a pool** is visible the same way: at 1 and 2 connections the database sat idle while callers queued.



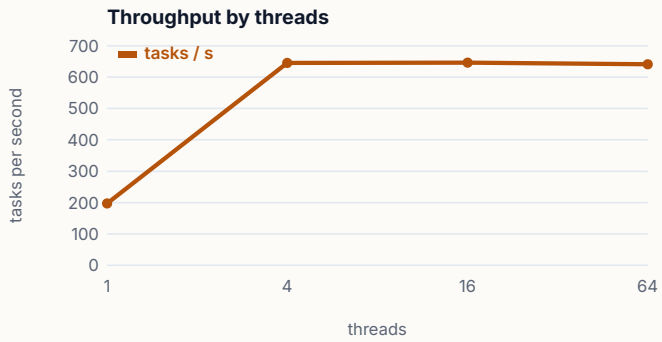


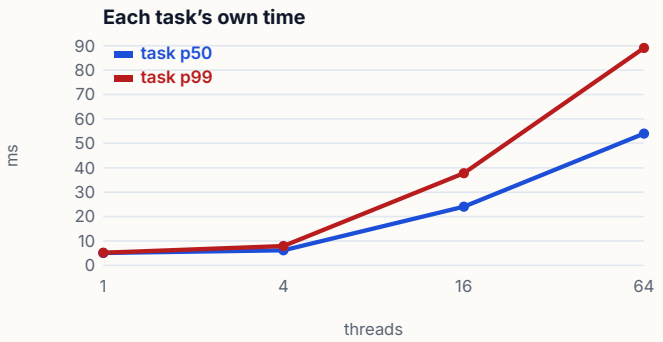
The same question with a real resource: 4 CPU cores

THREADS	TOTAL MS	TASKS/S	TASK P50 MS	TASK P99 MS
1	2032	197	5.0	5.2
4	620	645	6.1	7.9
16	619	646	24.0	37.8
64	624	641	54.0	89.1

400 tasks of about 5 ms of arithmetic each · real output · GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1

- **Throughput is capped by cores:** 197 tasks/s on 1 thread, 645 on 4, 641 on 64 — no gain past the core count.
- **Each task got slower** as threads were added: p50 5.0 ms alone, 54.0 ms with 64 threads, because the cores were shared. A timeout set from a single-thread test would fail under load.
- **For I/O-bound work** the right size is set by the resource behind it (the database's slots), for CPU-bound work by the cores. Measure both throughput and per-task latency when you change it.



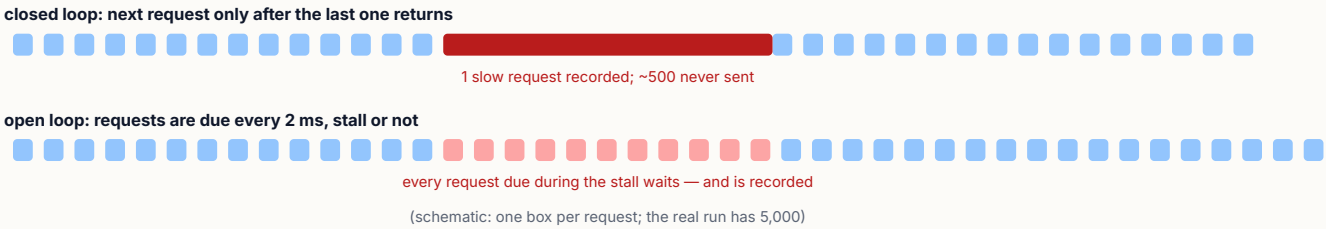


SOURCES results: deepdive/logs/pool.txt, cpupool.txt · deepdive/src/.../pool/PoolSize.java, CpuPool.java

05 · MEASURING LATENCY

Coordinated omission

Most load tests send a request, wait for the answer, then send the next. When the server stalls, the test stalls with it — and stops sending the requests that real users would have kept sending. Those requests are never measured. This is **coordinated omission**, the term used in the wrk2 README and the HdrHistogram Javadoc: the tester slows down with the system it measures, and omits the requests that would have seen the stall.



(schematic: one box per request; the real run has 5,000)

Two ways to get the truth

Measure from the due time. Keep a schedule — an *open* workload — and count each request's latency from when it should have been sent. wrk2 "measures response latency from the time the transmission *should* have occurred".

Or correct afterwards. HdrHistogram's `recordValueWithExpectedInterval` adds, for a value longer than the expected interval, "an additional series of decreasingly-smaller … value records" — the requests that would have queued behind it.

Open and closed models in load tools

> "In a closed model, the execution time of each iteration dictates the number of iterations executed … In the open model, on the other hand, VUs arrive independently of iteration completion." — k6 documentation

k6: `constant-arrival-rate`, `ramping-arrival-rate` (open). Gatling: `constantUsersPerSec`, `rampUsersPerSec` (open) versus `constantConcurrentUsers` (closed). To reproduce users who arrive independently of each other's responses, use an open model.

The arithmetic of this chapter's experiment

The program (constants read from `CoordinatedOmission.java`) sends one request every 2 ms for 10 s — 5,000 due — and the server stalls once, for 1000 ms. **Open loop:** 500 requests fall due during the stall, 10% of all; they wait from about 1000 ms down to 0, so everything from about p90 up is stall. **Closed loop:** the client waits with the server, so only one request sees the stall — 1 of 4,501 samples, 0.02%: it can only appear at p99.99 or the maximum. Chapter 5's second page shows exactly that.

Percentiles, read correctly

Stats.java — the percentile every experiment here uses

```
/** Nearest-rank percentile, in milliseconds. */
public double ms(double percentile) {
  int rank =
      (int) Math.ceil(percentile / 100 * values.length);
  return values[Math.max(rank, 1) - 1] / 1e6;
}
```

p99 of 5,000 samples is the 4,950th smallest — the 50 slowest are above it. Nearest-rank never interpolates: the value reported was really measured.

REQUESTS PER PAGE VIEW	CHANCE ONE OF THEM IS SLOWER THAN P99
1	1%
10	10%
50	39%
100	63%
200	87%

The table is arithmetic, not a measurement: $1 - 0.99^n$. A page that makes 100 backend calls shows a p99 delay to most of its users — so for the calls behind a page, p99.9 and the maximum are the numbers that describe the user's experience.

SOURCES HdrHistogram AbstractHistogram.recordValueWithExpectedInterval (Javadoc) · wrk2 README (github.com/giltene/wrk2) · k6 docs, "Open and closed models" · Gatling docs, "Injection"

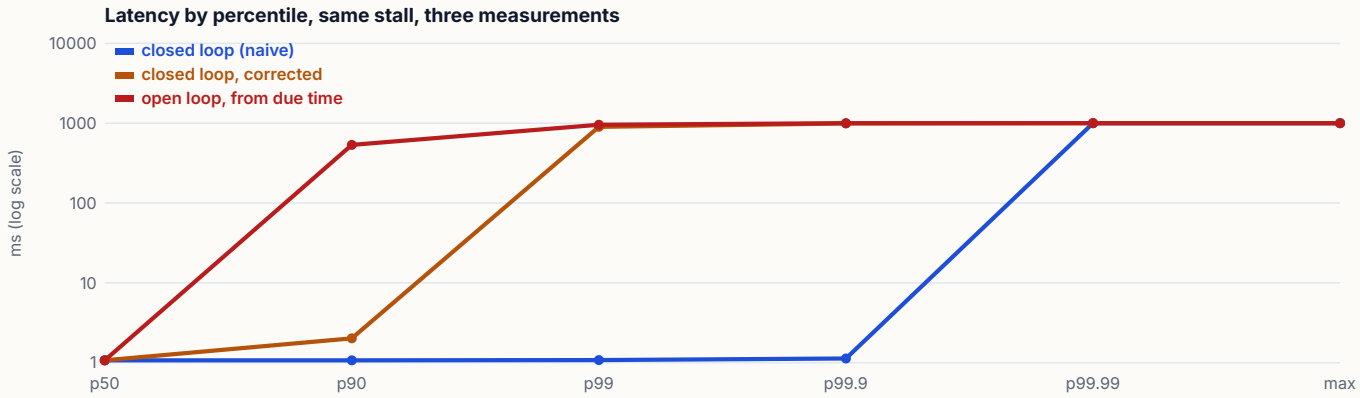
05 · MEASURING LATENCY

Measured: closed loop, corrected, open loop

A service that answers in 1 ms, except once: from second 2 to second 3 of a 10-second run it stalls. The intended rate is one request every 2 ms (500 per second). The same stall, measured three ways:

MEASUREMENT	SAMPLES	P50 MS	P90 MS	P99 MS	P99.9 MS	P99.99 MS	MAX MS
closed loop — what most tests report	4501	1.07	1.07	**1.08**	1.13	1000.14	1000.14
closed loop, HdrHistogram-style correction	5000	1.07	2.02	900.14	992.14	1000.14	1000.14
open loop, latency from the due time	5000	1.07	534.72	**954.10**	996.53	1000.14	1000.14

Real output · GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1



open loop: latency from the due time

```
/** Requests are due every 2 ms; latency from due time. */
static long[] openLoop() throws Exception {
  int total = (int) (RUN_NS / INTERVAL);
  long[] out = new long[total];
  start = System.nanoTime();
  for (int i = 0; i < total; i++) {
    long due = start + i * INTERVAL;
    while (System.nanoTime() < due) {
      Thread.onSpinWait();
    }
    serve();
    out[i] = System.nanoTime() - due;
  }
  return out;
}
```

the correction, as HdrHistogram does it

```
/**
 * HdrHistogram's recordValueWithExpectedInterval: for a value
 * longer than the interval, also record the requests that
 * would have been sent meanwhile: value - k·interval.
 */
static long[] corrected(long[] values) {
  long[] out = new long[values.length * 2];
  int n = 0;
  for (long v : values) {
    out = grow(out, n);
    out[n++] = v;
    for (long m = v - INTERVAL; m ≥ INTERVAL;
        m -= INTERVAL) {
      out = grow(out, n);
      out[n++] = m;
    }
  }
  return Arrays.copyOf(out, n);
}
```

- **The naive p99 was 1.08 ms**. Measured from the due times, p99 was **954.10 ms**: one request in ten was due during the stall, and more waited while the backlog drained.
- **The naive run recorded 4501 samples, not 5,000**: the missing 499 are the requests the client did not send while it waited. The max was right; the percentiles below it were not.
- **The correction recovered most of it** (p99 900.14 ms) from the closed-loop data alone, because the expected interval was known.
- **Report max and high percentiles, with the sample count.** A drop in the count during a test is a stall the percentiles cannot show.
- **Generate load on a schedule** (open model) and measure from the due time; if your tool cannot, correct with the known interval.
- **Percentiles cannot be averaged** across machines or intervals: merge the recorded values (or histograms), then read the percentile.

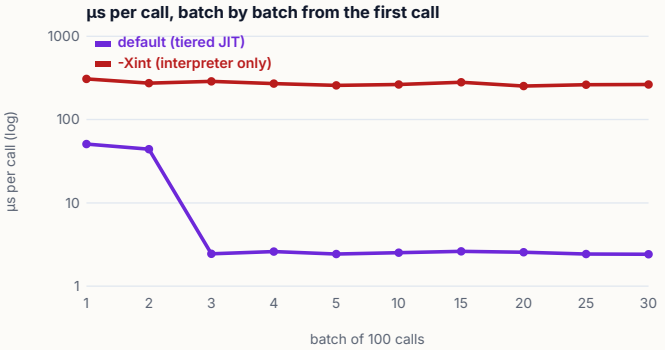
SOURCES results: deepdive/logs/co.txt · deepdive/src/.../latency/CoordinatedOmission.java · G. Tene, “How NOT to Measure Latency” (conference talk)

06 · JIT AND BENCHMARKS

Tiered compilation and warm-up

HotSpot starts every method in the interpreter, counts calls and loop iterations, and compiles the hot ones — first quickly with C1, collecting a profile, then again with C2 using that profile. Measurements taken before the final version is installed include slower code: in the run below, the first two batches.

LEVEL	CODE	USED FOR
0	interpreter	runs bytecode; profiles the method
1	C1, full optimisation, no profiling	trivial methods
2	C1 with invocation and back-edge counters	when C2 is busy
3	C1 with full profiling	the usual first compiled form
4	C2, profile-guided optimisation	the fast steady state



On the runner's JVM: `Tier3InvocationThreshold` = 200, `Tier4InvocationThreshold` = 5000, `Tier4BackEdgeThreshold` = 40000, `CICompilerCount` = 3 — from `-XX:+PrintFlagsFinal`. They are not exact call counts: `compilationPolicy.hpp` compares the counters with each threshold multiplied by a scale factor that depends on the compile queues.

WarmUp.java — 30 batches of 100 calls

```
for (int batch = 1; batch <= 30; batch++) {
  long t0 = System.nanoTime();
  for (int i = 0; i < 100; i++) {
    sink = sumOfSquares(DATA);
  }
  double us = (System.nanoTime() - t0) / 1e3 / 100;
  if (batch <= 5 || batch % 5 == 0) {
    System.out.printf("%5d %10.2f%n", batch, us);
  }
}
```

-XX:+PrintCompilation · real output (ms · id · % OSR · level)

```
33   11 %     3       deep.jit.WarmUp::sumOfSquares @ 11 (42
bytes)
33   12       3       deep.jit.WarmUp::sumOfSquares (42
bytes)
33   13 %     4       deep.jit.WarmUp::sumOfSquares @ 11 (42
bytes)
38   11 %     3       deep.jit.WarmUp::sumOfSquares @ 11 (42
bytes)   made not entrant
72  122       4       deep.jit.WarmUp::sumOfSquares (42
bytes)
75   12       3       deep.jit.WarmUp::sumOfSquares (42
bytes)   made not entrant
```

Reading a PrintCompilation line

COLUMN	MEANING
1	milliseconds since the JVM started
2	compile id — the same id for every event of one compilation
3	flags: `%` on-stack replacement · `s` synchronized · `!` has exception handlers · `b` blocking compile · `n` native wrapper
4	tier, 0–4 (table above)
5	the method; `@ 11` after an OSR compile is the bytecode index of the loop it enters at; then the bytecode size
…	`made not entrant`: the method's entry is patched so new calls no longer enter this code (`nmethod::make_not_entrant`). In the output above it follows the level 4 versions.

- **Interpreter only:** about 265 µs per call. **Compiled:** about 2.5 µs — 105× faster, the same code.
- **The first batches** ran at 51–51 µs while compilation was under way.
- **Exclude the warm-up from a measurement**, or report it separately: keep measuring until the values stop changing. JMH runs warm-up iterations (`-wi`) before measured ones and forks a fresh JVM (`-f`).
- **In a load test, plot latency over time** rather than one percentile for the whole run, so the warm-up is visible.

SOURCES OpenJDK 21 hotspot/share/compiler/compilationPolicy.hpp (the five levels), compileTask.cpp (the print format), code/nmethod.cpp (make_not_entrant) · -XX:+PrintCompilation, -XX:+PrintFlagsFinal on the runner · results: deepdive/logs/warmup*.txt, jit-flags.txt

06 · JIT AND BENCHMARKS

Dead-code elimination and constant folding, measured with JMH

The JIT may remove code whose result is never used (dead-code elimination) and compute expressions on constants ahead of time (constant folding). A hand-written timing loop cannot tell; JMH can — compare with an empty baseline and hand results back.

DeadCodeBench.java (1/2)

```java
@State(Scope.Thread)
@BenchmarkMode(Mode.AverageTime)
@OutputTimeUnit(TimeUnit.NANOSECONDS)
public class DeadCodeBench {
  static final double CONSTANT = Math.PI;
  static final int[] FIXED = new int[1_000];
  double x = Math.PI;
  int[] data = new int[1_000];

  @Benchmark
  public void baseline() {
    // nothing: the cost of the harness itself
  }

  @Benchmark
  public void discarded() {
    Math.log(x);                    // result unused: dead code
  }

  @Benchmark
  public double returned() {
    return Math.log(x);           // JMH consumes the result
  }

  @Benchmark
  public void blackhole(Blackhole bh) {
    bh.consume(Math.log(x));
  }

  @Benchmark
  public double constant() {
    return Math.log(CONSTANT);    // foldable at compile time?
  }
```

DeadCodeBench.java (2/2)

```java
  @Benchmark
  public void sumDiscarded() {
    long s = 0;
    for (int v : data) {
      s += v;                       // the sum is never used
    }
  }

  @Benchmark
  public long sumReturned() {
    long s = 0;
    for (int v : data) {
      s += v;
    }
    return s;
  }

  @Benchmark
  public void mulDiscarded() {
    double r = x * x + 1;           // never used
  }

  @Benchmark
  public double mulReturned() {
    return x * x + 1;
  }

  @Benchmark
  public double mulConstant() {
    return CONSTANT * CONSTANT + 1;  // static final: foldable
  }
}
```

BENCHMARK	NS/OP	ERROR (99.9%)	MINUS BASELINE
`baseline`	0.371	± 0.003	+0.000
`blackhole`	7.092	± 0.039	+6.721
`constant`	7.001	± 0.022	+6.630
`discarded`	7.102	± 0.029	+6.731
`mulConstant`	0.625	± 0.003	+0.254
`mulDiscarded`	0.375	± 0.022	+0.004
`mulReturned`	0.627	± 0.007	+0.256
`returned`	7.084	± 0.009	+6.713
`sumDiscarded`	0.710	± 0.011	+0.339
`sumReturned`	105.959	± 0.797	+105.588

JMH 1.37 · 1 fork, 3 warm-up and 5 measured iterations of 1 s · GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1

- **The empty baseline costs 0.37 ns**: the harness itself. Anything at that level did no work.
- **Math.log, result unused**: 7.10 ns; returned 7.08 ns; into a Blackhole 7.09 ns — **not** removed: the unused call still ran.
- **Math.log of a static final constant**: 7.00 ns — **not** folded here: it cost the same as the call on a field.
- **A pure loop summing 1,000 ints**: 0.71 ns unused, 105.96 ns returned — the JIT **removed** the whole loop.
- `x * x + 1`: 0.38 ns unused, 0.63 ns returned, 0.62 ns on constants — the JIT **removed** the arithmetic.

- **Never trust a benchmark you have not seen fail:** add a baseline and a variant that should be eliminated.
- **Return results or use a Blackhole**, read inputs from `@State` fields, and let JMH fork a fresh JVM (`-f`).

SOURCES JMH 1.37 (org.openjdk.jmh): Blackhole, @State, -f/-wi/-i options · results: deepdive/logs/jmh.txt, jmh-full.txt

06 · JIT AND BENCHMARKS

Inside C2: inlining, receiver types, escape analysis

Most of C2's speed comes from three things it does with the interpreter's profile: copying small called methods into the caller (inlining), betting on the receiver types a call site has seen, and removing allocations of objects that never leave a method (escape analysis). Each is controlled by flags; the values below are the runner's.

A call site's receiver types — `doCall.cpp`

The profile records up to `TypeProfileWidth` receiver classes per call site. C2 then generates:

- **1 type**: a type check and the inlined method; another type takes an *uncommon trap* and the method is deoptimized.
- **2 types** (`UseBimorphicInlining`): two checks, both methods inlined.
- **More**: the most frequent type inlined only if it reaches `TypeProfileMajorReceiverPercent` of calls; otherwise a virtual call.

`-XX:+PrintCompilation -XX:+PrintInlining · CallSites.java (real output, start)`

```
31  8 %  b  deep.jit.CallSites::oneType @ 10 (40 bytes)
@ 25  deep.jit.CallSites$A::area (2 bytes)  inline (hot)
35  9  b  deep.jit.CallSites::oneType (40 bytes)
@ 25  deep.jit.CallSites$A::area (2 bytes)  inline (hot)
37  10 %  b  deep.jit.CallSites::fourTypes @ 10 (40 bytes)
@ 25  deep.jit.CallSites$Shape::area (0 bytes)  virtual call
39  11  b  deep.jit.CallSites::fourTypes (40 bytes)
@ 25  deep.jit.CallSites$Shape::area (0 bytes)  virtual call
```

Run with C2 only (`-XX:-TieredCompilation`), so each inlining line follows its own method's compile line. In `oneType` the call at bytecode 25 became the inlined `A::area` ; in `fourTypes` the same call stayed a `virtual call` to `Shape::area` .

FLAG	RUNNER	DESCRIPTION IN C2_GLOBALS.HPP / GLOBALS.HPP
`MaxInlineSize`	35	"The maximum bytecode size of a method to be inlined by high tier compiler"
`FreqInlineSize`	325	"The maximum bytecode size of a frequent method to be inlined"
`InlineSmallCode`	2500	"Only inline already compiled methods if their code size is less than this"
`MaxInlineLevel`	15	"maximum number of nested calls that are inlined by high tier compiler"
`TypeProfileWidth`	2	"Number of receiver types to record in call/cast profile"
`TypeProfileMajorReceiverPercent`	90	"% of major receiver type to all profiled receivers"
`DoEscapeAnalysis`	true	"Perform escape analysis"
`EliminateAllocations`	true	"Use escape analysis to eliminate allocations"

Measured

BENCHMARK	NS/OP	ERROR	BYTES ALLOCATED / OP
tiny static method, inlined	0.598	± 0.011	0
same method, `DONT_INLINE`	1.872	± 0.024	0
call site saw 1 type (1,024 calls)	297.269	± 0.963	0
call site saw 2 types	568.492	± 3.845	0
call site saw 4 types	17021.088	± 2759.912	0
`new Point` that never escapes	0.588	± 0.010	0
same, `-XX:-DoEscapeAnalysis`	2.484	± 0.008	24

InliningBench.java · JMH 1.37 with `-prof gc` (`gc.alloc.rate.norm`) · each benchmark in its own JVM · GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1

- **Not inlining a two-int addition** cost 1.27 ns per call.
- **The call site with 4 receiver types** took 57.3× as long as with 1 (17021 against 297 ns for 1,024 calls); 2 types: 568 ns.
- **Escape analysis**: the non-escaping `Point` allocated 0 bytes per call; with it switched off, 24.

SOURCES OpenJDK 21 hotspot/share/opto/c2_globals.hpp, runtime/globals.hpp, opto/doCall.cpp (Compile::call_generator) · JMH 1.37 CompilerControl, GCProfiler · results: deepdive/logs/jmh-inline.txt, jit-inlining.txt, jit-flags.txt

07 · GC PAUSES AND TAIL LATENCY

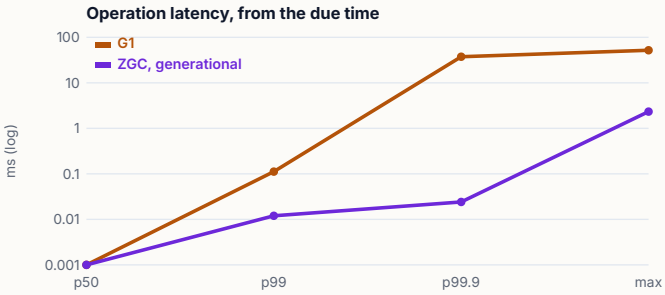
GC pauses: G1 and generational ZGC

A garbage collector has to find live objects and make room. Some of that work needs every application thread stopped — a **pause**. G1, the default on a server-class machine, copies live objects during its young and mixed pauses. ZGC copies them concurrently; Oracle's JDK 21 GC tuning guide states "max pause times under a millisecond … Pause times are independent of heap size".

The collectors, as of JDK 21–24

G1: default when the JVM sees at least 2 CPUs and about 1792 MB (chapter 9). Young and mixed collections copy live objects during pauses; marking of the old generation runs concurrently.

Generational ZGC: JEP 439, JDK 21, enabled with `-XX:+UseZGC -XX:+ZGenerational`; the default ZGC mode from JDK 23 (JEP 474); the only mode from JDK 24 (JEP 490). The tuning guide: "at the cost of some throughput".



A G1 cycle, as it happened in the measured window

PAUSE	COUNT	LONGEST MS	EVACUATION FAILED	WHAT IT DOES (JDK 21 G1 TUNING GUIDE)
Young (Normal)	5	9.706	–	copies live young objects; promotes survivors to old
Young (Concurrent Start)	2	8.782	–	a young collection that also starts concurrent marking
Remark	2	0.613	–	finishes marking, processes references, frees empty regions
Cleanup	2	0.308	–	decides whether mixed collections follow
Young (Prepare Mixed)	2	8.564	1	the last young-only collection before mixed ones
Young (Mixed)	3	13.822	2	young regions plus some old regions, copied
Full	1	46.387	–	fallback: compacts the whole heap in place, all threads stopped

Evacuation failure (3 of these pauses): G1 could not find free regions to copy live objects into. The tuning guide: G1 then hopes marking and mixed collections catch up; "if this assumption doesn't hold, then G1 will eventually schedule a Full GC" — which is what happened here: one full, in-place compaction of the heap (1023M→407M).

The workload

GcLatency.java — one operation due every 50 µs

```
for (int i = 0; i < total; i++) {
  long due = start + i * INTERVAL_NS;
  while (System.nanoTime() < due) {
    Thread.onSpinWait();
  }
  live[random.nextInt(LIVE)] = new byte[1024];
  sink = new byte[8 * 1024];
  latency[i] = System.nanoTime() - due;
}
```

A live set of 400,000 × 1 KB arrays (≈ 400 MB in a 1 GB heap) that keeps changing — every operation replaces one, so old objects keep dying — plus 8 KB of garbage per operation. One operation is due every 50 µs for 15 s; latency counts from the due time, the open-loop measurement from chapter 5. Pauses come from each JVM's own log (`-Xlog:gc,gc+phases`), from the moment measuring started.

SOURCES JEP 439 (JDK 21), JEP 474 (JDK 23), JEP 490 (JDK 24) · Java SE 21 HotSpot GC Tuning Guide: "Garbage-First (G1) Garbage Collector", "Available Collectors" · OpenJDK 21 gcConfig.cpp, os.cpp (default selection) · run: GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1, -Xms1g -Xmx1g

07 · GC PAUSES AND TAIL LATENCY

Measured: latency and pauses under one workload

COLLECTOR	OPERATIONS	P50 MS	P99 MS	P99.9 MS	MAX MS	PAUSES	LONGEST PAUSE MS	PAUSE TOTAL MS
G1	300,000	0.001	0.112	37.353	52.082	17	46.387	140.9
ZGC	300,000	0.001	0.012	0.024	2.341	22	0.013	0.2

Real output and GC logs · GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1

- **The median was the same** (0.001 ms against 0.001 ms): most operations never meet a pause.
- **The tail was not:** G1's p99.9 was 37.4 ms and its worst operation waited 52.1 ms; with ZGC p99.9 was 0.024 ms and the worst 2.3 ms.
- **The GC logs agree:** G1's longest pause was 46.4 ms; ZGC paused 22 times, the longest 0.013 ms. ZGC's worst operation (2.3 ms) was therefore not caused by a GC pause; the GC log does not show what delayed it.
- **Measure throughput as well as latency:** the tuning guide describes ZGC's low pauses as coming "at the cost of some throughput".
- **Log GC during load tests** (`-Xlog:gc`, as here) and compare pause times with the slow requests' times.
- **Size the heap on purpose** (`-Xmx` or `MaxRAMPercentage`): the default is 25% of the memory the JVM sees, which in a container is the container's limit.

Why p99 sits on the pauses

In an open loop, every operation that falls due during a pause waits for it. G1 was paused for 140.9 ms of the 15 s window: about 2,819 operations (one per 50 µs) were due while the application was stopped — 0.94% of all operations. Anything above that share is inside the tail: p99 is the slowest 1%, p99.9 the slowest 0.1%.

ZGC was paused for 0.21 ms in total — about 4 operations, 0.0014%: too few to reach p99.9.

FULL GC PHASE	MS
Mark live objects	13.627
Prepare compaction	4.048
Adjust pointers	5.120
Compact heap	21.589
Reset Metadata	0.839
pause	**46.387**

The full collection marked and compacted the whole heap while every application thread waited. Without it, G1's longest pause was 13.8 ms.

Every pause in the measured window

G1: 17 pauses

```
  8.437 ms  Young (Normal)
  3.546 ms  Young (Normal)
  5.678 ms  Young (Normal)
  9.457 ms  Young (Normal)
  9.706 ms  Young (Normal)
  8.782 ms  Young (Concurrent Start)
  0.613 ms  Remark
  0.289 ms  Cleanup
  8.564 ms  Young (Prepare Mixed)
 12.858 ms  Young (Mixed)
 13.822 ms  Young (Mixed)
  6.849 ms  Young (Mixed)
  2.232 ms  Young (Concurrent Start)
  0.583 ms  Remark
  0.308 ms  Cleanup
  2.817 ms  Young (Prepare Mixed)
 46.387 ms  Full
```

ZGC: 22 PAUSES	COUNT	LONGEST MS
Mark Start · minor	4	0.011
Mark End · minor	4	0.012
Relocate Start · minor	4	0.007
Mark Start · major, young	2	0.013
Mark End · major, young	2	0.011
Relocate Start · major, young	2	0.007
Mark End · major, old	2	0.013
Relocate Start · major, old	2	0.007

In this log, cycles tagged `y:` have only young-generation pauses; cycles tagged `Y:` also have `O:` (old) pauses and start with `Mark Start (Major)`. ZGC's driver runs minor collections of the young generation and major collections of both (JDK 21 `zDriver.cpp`).

SOURCES Java SE 21 HotSpot GC Tuning Guide (G1 cycle, evacuation failure) · OpenJDK 21 gc/z/zDriver.cpp (minor and major collections) · results: deepdive/logs/gc-G1.txt, gc-G1.log, gc-ZGC.txt, gc-ZGC.log

07 · GC PAUSES AND TAIL LATENCY

Inside the collectors: regions, cards and barriers

G1: regions and remembered sets

The heap is divided into equal regions. Without `G1HeapRegionSize`, the size is the heap divided by 2,048, kept between 1 MB and 32 MB and rounded up to a power of two (`HeapRegion::setup_heap_region_size`). An object of at least half a region is *humongous* and gets regions of its own (`humongous_threshold_for`).

To collect some regions without scanning the rest, "each heap region has a rem set that records the external heap references into it" (`g1RemSet.hpp`). After a reference store, G1's write barrier marks the 512-byte card (`GCCardSizeInBytes`) that holds the updated field; the cards are later processed into the remembered sets. Before the store, a second barrier logs the old value for snapshot-at-the-beginning marking (`g1BarrierSet.hpp`).

ZGC: coloured pointers and load barriers

Generational ZGC keeps metadata in the low bits of each stored reference (`zAddress.hpp`): remapped, marked-young, marked-old, finalizable and remembered bits. Its load barrier, from `zBarrier.hpp`:

> "The load barriers of ZGC check if a loaded value is safe to expose or not, and then shifts the pointer to remove metadata bits, such that it points to mapped memory."

In compiled code the check and the shift are one shift instruction; a pointer with a bad colour takes a slow path that fixes it — for example, finds the object's new address after relocation. This is how objects can move while the application runs.

Measured: region size

HEAP (-XMX)	G1 REGION SIZE
-Xmx1g	1M
-Xmx4g	2M
-Xmx16g	8M

`-Xlog:gc+init`

Measured: the cost of barriers

4,096 ACCESSES (MS)	PARALLEL	G1	GENERATIONAL ZGC
loadRefs	1.78	1.78	1.96
storeRefs	6.70	9.07	4.25
storeInts	2.85	2.93	2.51

BarrierBench.java · JMH, `-Xms1g -Xmx1g` and the collector's flag · GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1

- **Reference loads**: 1.96 µs with ZGC's load barrier against 1.78 (G1) and 1.78 (Parallel), which have none.
- **Reference stores**: 6.70 (Parallel), 9.07 (G1), 4.25 µs (ZGC). Int stores, which no collector intercepts: 2.51–2.93 µs.
- **The collectors differ in more than barriers** — heap layout and object placement too — so these numbers describe this loop on this machine, not a barrier cost in isolation.

Measured: pauses with a bigger heap

COLLECTOR	HEAP	P99.9 MS	LONGEST PAUSE MS	FULL GCS
G1	1 GB	37.353	46.387	1
G1	4 GB	5.144	10.853	0
ZGC	1 GB	0.024	0.013	0
ZGC	4 GB	0.017	0.012	0

Chapter 7's workload (≈ 400 MB live) with a 1 GB and a 4 GB heap.

SOURCES OpenJDK 21 gc/g1/heapRegion.cpp, heapRegionBounds.hpp, g1CollectedHeap.hpp, g1RemSet.hpp, g1BarrierSet.hpp; gc/shared/gc_globals.hpp (GCCardSizeInBytes); gc/z/zAddress.hpp, zBarrier.hpp · results: deepdive/logs/gc-init.txt, jmh-barrier.txt, gc-G1-4g.*, gc-ZGC-4g.*

08 · VIRTUAL THREADS

Virtual threads, carriers and pinning

A virtual thread (JEP 444, final in JDK 21) is a Java thread that is not tied to an OS thread. It runs *mounted* on a carrier — a platform thread of a small pool — and when it blocks on a socket, a sleep or a `java.util.concurrent` lock, it *unmounts* and the carrier runs another one.

Carriers

The scheduler is a `ForkJoinPool` whose parallelism is `availableProcessors()` unless `jdk.virtualThreadScheduler.parallelism` says otherwise (JDK 21 `VirtualThread.java`). The runner had 4.

Pinning (JDK 21–23)

Blocking inside `synchronized` keeps the virtual thread **pinned**: it cannot unmount, so it holds its carrier while it waits. JEP 491 (JDK 24) lets virtual threads unmount inside `synchronized`; the `jdk.tracePinnedThreads` property used here was removed with it.

VirtualThreads.java — the same wait, two locks

```
static void blockInSynchronized() {
  Object lock = new Object();             // one lock per task
  synchronized (lock) {
    sleep(100);                           // pins the carrier
  }
}

static void blockInLock() {
  var lock = new ReentrantLock();
  lock.lock();
  try {
    sleep(100);                           // unmounts
  } finally {
    lock.unlock();
  }
}
```

RUN	WALL TIME
10,000 tasks, 200 platform threads	5.04 s
10,000 tasks, virtual thread each	0.23 s
100 tasks, synchronized + sleep	2.50 s
100 tasks, ReentrantLock + sleep	0.10 s

Real output · JDK 21 · GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1

-Djdk.tracePinnedThreads=short · real output

```
VirtualThread[#10227]/runnable@ForkJoinPool-1-worker-3
reason:MONITOR

deep.vt.VirtualThreads.blockInSynchronized(VirtualThreads.ja
va:51) ⟸ monitors:1
```

- **10,000 blocking tasks:** 5.04 s on 200 platform threads (10,000 / 200 × 0.1 s = 5 s), 0.23 s with a virtual thread each.
- **Pinned:** 100 tasks holding a monitor while sleeping took 2.50 s = 100 / 4 carriers × 0.1 s. With `ReentrantLock`: 0.10 s.
- **On JDK 21–23, replace `synchronized` around blocking calls** with `java.util.concurrent` locks, or upgrade: JDK 24 removes this kind of pinning.
- **CPU-bound work is still limited by the cores:** the carriers number `availableProcessors()` (4 here), as in chapter 4's CPU pool.

What a blocking call does to the carrier, JDK 21

THE VIRTUAL THREAD BLOCKS IN…	CARRIER	EVIDENCE
`Thread.sleep`, `LockSupport.park`, `java.util.concurrent` locks and queues, socket I/O	**released** — the virtual thread unmounts	measured: 0.10 s above; sockets: `NioSocketImpl.park` → `Poller.poll`
`Object.wait()`, file I/O such as `FileInputStream.read`	held, but **compensated**: the pool adds a carrier for the duration	JDK 21 `Object.wait` and `FileInputStream.read` call `Blocker.begin()` → `ForkJoinPool` compensated block
any of the above inside `synchronized`	**pinned**, not compensated: the carrier waits too	measured above: 2.50 s = 100 / 4 × 0.1 s; `VirtualThread.parkOnCarrierThread`
a native method or foreign function on the stack	**pinned**	`Continuation.Pinned.NATIVE`

Compensation has a ceiling: `jdk.virtualThreadScheduler.maxPoolSize`, by default the larger of the parallelism and 256 (JDK 21 `VirtualThread.createDefaultScheduler`).

Finding pins in production: the JFR event `jdk.VirtualThreadPinned` is on in the default JFR settings, with a 20 ms threshold and a stack trace (JDK 21 `default.jfc`).

SOURCES JEP 444 (JDK 21), JEP 491 (JDK 24) · OpenJDK 21 java/lang/VirtualThread.java, Object.java, io/FileInputStream.java, jdk/internal/misc/Blocker.java, jdk/internal/vm/Continuation.java, sun/nio/ch/NioSocketImpl.java · jdk.jfr default.jfc · results: deepdive/logs/vt.txt

08 · VIRTUAL THREADS

Where a blocked thread keeps its stack

A platform thread is an OS thread with its own stack, reserved when the thread starts. A virtual thread that blocks is unmounted: its frames are copied off the carrier's stack into an object on the Java heap, and copied back when it continues. Both are visible from outside the program.

Platform thread: `ThreadStackSize`

On Linux x86-64 the default Java thread stack is **1024 KB** (`globals_linux_x86.hpp` : `define_pd_global(intx, ThreadStackSize, 1024)`); the runner reports `ThreadStackSize = 1024` . Native Memory Tracking counts it as *reserved*; only pages the thread touches are *committed*.

OpenJDK 21 java.base jdk/internal/vm/StackChunk.java (complete, without the licence header)

```
package jdk.internal.vm;

public final class StackChunk {
    public static void init() {}

    private StackChunk parent;
    private int size;    // in words
    private int sp;      // in words
    private int bottom;  // in words

    // The stack itself is appended here by the VM, as well as some injected fields

    public StackChunk parent() { return parent; }
    public boolean isEmpty()   { return sp == bottom; }
}
```

Measured: 10,000 blocked threads

10,000 THREADS BLOCKED 20 FRAMES DEEP	PLATFORM	VIRTUAL
OS threads (NMT)	10,022	27
thread stacks reserved (NMT)	10,054 MB	26 MB
thread stacks committed (NMT)	376 MB	3 MB
`StackChunk` **objects on the heap**	0	10,000
their size	0 MB	27 MB
heap used after GC	4 MB	31 MB
process resident memory (VmRSS)	315 MB	92 MB

ThreadMemory.java — each thread blocks 20 frames deep

```
static void deep(int depth) {
  if (depth > 0) {
    deep(depth - 1);
    return;
  }
  blocked.countDown();
  try {
    release.await();
  } catch (InterruptedException e) {
    Thread.currentThread().interrupt();
  }
}
```

ThreadMemory.java with `-XX:NativeMemoryTracking=summary` ; it runs `jcmd VM.native_memory summary` and `jcmd GC.class_histogram` on itself · GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1

- **Platform threads**: 10,022 OS threads, 10,054 MB of stack address space reserved — 1,027 KB each, the `ThreadStackSize` of 1024 KB — of which 376 MB committed.
- **Virtual threads**: 10,000 `StackChunk` objects, 2,797 bytes each on average, on the Java heap; the process had 27 OS threads.
- **Resident memory**: 315 MB against 92 MB.

SOURCES OpenJDK 21 hotspot/os_cpu/linux_x86/globals_linux_x86.hpp (ThreadStackSize), java.base jdk/internal/vm/StackChunk.java, Continuation.java · Native Memory Tracking (services/nmtDCmd.hpp) · results: deepdive/logs/vt-memory.txt

09 · CONTAINERS AND CPUS

How the JVM reads cgroup limits

A container's limits are cgroup settings: a CPU **quota** per period (`cpu.max`) and a memory limit (`memory.max`). With `UseContainerSupport`, true by default in JDK 21 (`globals_linux.hpp`), the JVM reads both at start-up and sizes itself from them — the number of CPUs it reports, the heap, the garbage collector, and every thread pool sized from `availableProcessors()`.

CPUs from the quota

`--cpus=N` writes `cpu.max` as *quota period*. HotSpot divides quota by period and **rounds up**, capped at the host's CPUs (JDK 21 `cgroupUtil_linux.cpp`: `ceilf(quota / period)`). `-XX:ActiveProcessorCount` overrides the result.

Heap from the memory limit

Without `-Xmx`, the maximum heap is `MaxRAMPercentage` of the memory the JVM sees — **25** by default (`gc_globals.hpp`). `MinRAMPercentage` (50) is used instead only when memory is so small that half of it is below the default `MaxHeapSize` (`arguments.cpp`).

The collector from both

G1 is chosen only on a "server-class machine": at least **2** active processors and at least **2 GB − 256 MB** of memory (`os.cpp` `is_server_class_machine`, with an extra check for hyper-threaded packages). Otherwise **Serial** — one thread, stop-the-world (`gcConfig.cpp`).

Measured: the same JVM, different limits

DOCKER RUN …	CPU.MAX	MEMORY.MAX	CPUS	MAX HEAP	HEAP / LIMIT	COMMON POOL	GC
`no limits`	max/100000	max	**4**	3,998 MB	–	3	**G1**
`--cpus=2 --memory=2g`	200000/100000	2,048 MB	**2**	512 MB	25.0%	1	**G1**
`--cpus=1 --memory=1g`	100000/100000	1,024 MB	**1**	247 MB	24.1%	1	**Serial**
`--cpus=0.5 --memory=512m`	50000/100000	512 MB	**1**	123 MB	24.0%	1	**Serial**
`--cpus=1 --memory=1g -XX:ActiveProcessorCount=4`	100000/100000	1,024 MB	**4**	247 MB	24.1%	3	**Serial**
`--cpus=1 --memory=1g -XX:MaxRAMPercentage=75`	100000/100000	1,024 MB	**1**	742 MB	72.5%	1	**Serial**

eclipse-temurin:21-jdk on GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1 · the program prints `/sys/fs/cgroup` and what the JVM decided · GC names: Copy + MarkSweepCompact is Serial

Resources.java

```
public static void main(String[] args) throws Exception {
  var rt = Runtime.getRuntime();
  String gcs = ManagementFactory.getGarbageCollectorMXBeans()
      .stream().map(b -> b.getName())
      .collect(Collectors.joining(", "));
  System.out.printf("cgroup cpu.max %-14s memory.max %-11s"
      + " | CPUs %d  max heap %4d MB  commonPool %d  GC %s%n",
      cgroup("cpu.max"), cgroup("memory.max"),
      rt.availableProcessors(), rt.maxMemory() / (1 << 20),
      ForkJoinPool.getCommonPoolParallelism(), gcs);
}
```

- **--cpus=0.5** is reported as **1** CPU: 0.5 rounded up. The JVM cannot express half a CPU; the kernel enforces it by throttling.
- **2 CPUs and 2 GB** crossed the server-class line: **G1**, heap 512 MB. **1 CPU and 1 GB** did not: **Serial**.
- **ActiveProcessorCount=4** made the JVM report 4 CPUs (common pool 3) — and it still chose **Serial**: 1 GB is below the memory line.
- **MaxRAMPercentage=75** raised the heap from 247 MB to 742 MB in the same 1 GB container.

A quota is time, not cores

`--cpus=0.5` lets the container run for half of each scheduling period, then stops it until the next one. The JVM still reports 1 CPU and runs one GC thread. A multi-threaded burst can use the whole quota early in the period and then wait — throttling, visible in `cpu.stat` as `nr_throttled` and `throttled_usec`.

SOURCES OpenJDK 21: os/linux/cgroupUtil_linux.cpp, os/linux/globals_linux.hpp, runtime/os.cpp, gc/shared/gcConfig.cpp, gc/shared/gc_globals.hpp, runtime/arguments.cpp · results: deepdive/logs/containers.txt

09 · CONTAINERS AND CPUS

Defaults sized from the CPU count; CPU throttling

These defaults are computed from `availableProcessors()`, which reported 1 for both `--cpus=1` and `--cpus=0.5` in the runs on the previous page.

SIZED FROM THE CPU COUNT	DEFAULT	SOURCE	MEASURED (HOST / 2 / 1 CPU)
`ForkJoinPool.commonPool()` — parallel streams, `CompletableFuture.*Async`	CPUs − 1, at least 1	JDK 21 `ForkJoinPool.java`	3 / 1 / 1
virtual-thread carriers	CPUs	JDK 21 `VirtualThread.java`	chapter 8
G1/Parallel GC worker threads (`ParallelGCThreads`)	CPUs up to 8, then 8 + 5/8 of the rest	JDK 21 `workerPolicy.cpp`	–
Netty event-loop threads	2 × CPUs	Netty 4.1 `MultithreadEventLoopGroup`	–
the garbage collector itself	Serial below 2 CPUs or 1792 MB	`os.cpp` , `gcConfig.cpp`	Serial at 1 CPU

The heap is not the whole process

`memory.max` is a "memory usage hard limit" for the whole cgroup; if usage reaches it and cannot be reduced, "the OOM killer is invoked" (cgroup-v2.rst). The JVM uses memory outside the heap too: Native Memory Tracking reports it under Class, Thread Stack, Code, GC, Compiler, Internal, Other and more (JDK 21 `allocation.hpp`). Raising `MaxRAMPercentage` to 75 (measured above: 742 MB of 1,024 MB) leaves the rest for that memory; measure it with `-XX:NativeMemoryTracking=summary` and `jcmd <pid> VM.native_memory` .

- **Run performance tests with the production limits** — the same `--cpus` and `--memory` . A laptop or an unlimited runner chooses a different collector, a different heap and different pool sizes.
- **Print what the JVM decided:** `-Xlog:gc` prints the collector at start-up ("Using G1" in chapter 7's log); `-XX:+PrintFlagsFinal` marks values the JVM chose itself `{ergonomic}` .
- **Give a JVM service at least 2 CPUs and 2 GB**, or choose the collector yourself (`-XX:+UseG1GC`) — otherwise a small pod gets Serial.
- **Set the heap on purpose** — `-Xmx` , or `MaxRAMPercentage` for a heap that follows the limit.

Measured: throttling

DOCKER RUN …	BUSY THREADS	LATE P50 MS	LATE P99 MS	LATE MAX MS	PERIODS	THROTTLED PERIODS
`no limit`	0	0.1	0.1	**0.2**	–	–
`--cpus=0.5`	0	0.1	0.1	**0.2**	29	0
`no limit`	3	0.1	0.4	**1.7**	–	–
`--cpus=2`	3	0.1	0.4	**30.8**	31	31
`--cpus=0.5`	3	0.1	84.8	**86.0**	42	42

Throttle.java: busy threads use the quota while a timer thread sleeps 5 ms, 600 times, and records how late it wakes; periods and throttling from the container's `cpu.stat` · GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1

- **--cpus=0.5 with 3 busy threads:** 42 of 42 periods throttled, and a 5 ms sleep woke up to **86 ms** late — against 1.7 ms with no limit.
- **Why the timer thread was late:** the busy threads used the period's quota, and `cpu.max` allows the cgroup "up to $MAX in each $PERIOD" (cgroup-v2.rst), so the timer thread could not run until the next period began.

SOURCES OpenJDK 21: java/util/concurrent/ForkJoinPool.java, java/lang/VirtualThread.java, gc/shared/workerPolicy.cpp, memory/allocation.hpp and services/nmtDCmd.hpp (Native Memory Tracking) · Netty 4.1 MultithreadEventLoopGroup.java · Linux 6.17 Documentation/admin-guide/cgroup-v2.rst (cpu.max, cpu.stat, memory.max) · results: deepdive/logs/containers.txt, throttle.txt

10 · THE JAVA MEMORY MODEL

Data races, happens-before and store buffering

Without synchronisation, Java promises much less than the source code suggests. The compiler and the CPU may reorder and cache memory accesses as long as a single thread cannot tell. The Java Memory Model (JLS chapter 17) says exactly when another thread is guaranteed to see a write — and that guarantee is the *happens-before* order.

Data race (JLS §17.4.5)

Two accesses to the same variable **conflict** if at least one is a write. A program has a **data race** when two conflicting accesses are not ordered by happens-before.

The guarantee (§17.4.3): "If a program has no data races, then all executions of the program will appear to be sequentially consistent." With a race, the JMM allows results no interleaving of the source lines can produce.

Happens-before, JLS §17.4.5 (quoted)

- An unlock on a monitor *happens-before* every subsequent lock on that monitor.
- A write to a `volatile` field *happens-before* every subsequent read of that field.
- A call to `start()` on a thread *happens-before* any actions in the started thread.
- All actions in a thread *happen-before* any other thread successfully returns from a `join()` on that thread.

Plus program order within a thread, and transitivity. The `java.util.concurrent` package documentation ("Memory Consistency Properties") extends them to concurrent collections, executors, futures, locks and synchronizers.

Measured with jcstress: store buffering

StoreBufferingTest.java (outcomes declared with @Outcome)

```
@State
public class StoreBufferingTest {
  int x;
  int y;

  @Actor
  public void first(II_Result r) {
    x = 1;
    r.r1 = y;
  }

  @Actor
  public void second(II_Result r) {
    y = 1;
    r.r2 = x;
  }
}
```

Plain fields — every interleaving of the four lines gives at least one 1. Yet:

RESULT	SAMPLES	SHARE	EXPECTED	MEANING
`0, 0`	281,351,791	43.51%	Interesting	both read before either write was visible
`0, 1`	195,672,763	30.26%	Acceptable	some interleaving of the four steps
`1, 0`	169,298,086	26.18%	Acceptable	some interleaving of the four steps
`1, 1`	368,708	0.06%	Acceptable	some interleaving of the four steps

The same with `volatile int x, y` — `0, 0` declared forbidden:

RESULT	SAMPLES	SHARE	EXPECTED	MEANING
`0, 0`	0	0.00%	Forbidden	both read before either write was visible
`0, 1`	292,059,994	54.63%	Acceptable	some interleaving of the four steps
`1, 0`	233,026,600	43.59%	Acceptable	some interleaving of the four steps
`1, 1`	9,548,434	1.79%	Acceptable	some interleaving of the four steps

- **281,351,791 of 646,691,348 plain runs (43.51%) ended 0, 0**: both threads read before either write was visible. No interleaving of the source allows that; the JMM does, because the program has a data race.
- **With volatile: 0 of 534,635,028**. Volatile reads and writes are synchronisation actions, totally ordered (§17.4.4), so one write must come first and the other thread's read must see it.
- **The result does not say whether the compiler or the CPU reordered the accesses.** The log shows jcstress ran the actors interpreted, with C1 and with C2, and with C2's stress options (`-XX:+StressLCM`, `-XX:+StressGCM` …); the JMM allows 0, 0 for this racy program in every case.
- **The machine:** 4 CPUs (AMD EPYC 7763 64-Core Processor); jcstress pins each actor thread to its own CPU.

SOURCES JLS SE 21 §17.4.2–17.4.5 · jcstress 0.16 (org.openjdk.jcstress) · OpenJDK 21 java/util/concurrent/package-info.java · results: ci-logs-ops logs/jcstress.log

10 · THE JAVA MEMORY MODEL

Visibility and lost updates

Happens-before is also about *whether* a write is seen at all. A thread spinning on a plain field has no happens-before edge with the thread that sets it, and JLS §17.3 allows the loop to keep a value it read once: “The compiler is free to read the field `this.done` just once, and reuse the cached value in each execution of the loop.”

StaleFlag.java — the loop runs 1 s, then the flag is set

```
static String stops(boolean useVolatile) throws Exception {
  Thread spinner = new Thread(() -> {
    if (useVolatile) {
      while (!marked) {
        // spin
      }
    } else {
      while (!plain) {
        // spin
      }
    }
  });
  spinner.setDaemon(true);
  spinner.start();
  Thread.sleep(1_000);      // let the JIT compile the loop
  if (useVolatile) {
    marked = true;
  } else {
    plain = true;
  }
  spinner.join(3_000);
  return spinner.isAlive()
      ? "still spinning 3 s after the flag was set"
      : "stopped";
}
```

real output · 5 runs in CI

```
plain field:    still spinning 3 s after the flag was set
volatile field: stopped
plain field:    still spinning 3 s after the flag was set
volatile field: stopped
plain field:    still spinning 3 s after the flag was set
volatile field: stopped
plain field:    still spinning 3 s after the flag was set
volatile field: stopped
plain field:    still spinning 3 s after the flag was set
volatile field: stopped
```

- **Plain field: 5 of 5 runs** still spinning 3 s after the flag was set, as §17.3 allows.
- **Volatile: 5 of 5** stopped. A write to a volatile field happens-before every subsequent read of that field (§17.4.5).

The same question, asked by jcstress

SpinTest.java — Mode.Termination

```
@JCStressTest(Mode.Termination)
@Outcome(id = "TERMINATED", expect = ACCEPTABLE,
    desc = "saw it")
@Outcome(id = "STALE", expect = ACCEPTABLE_INTERESTING,
    desc = "spins forever")
@State
public class SpinTest {
  boolean ready;

  @Actor
  public void spin() {
    while (!ready) {
      // wait
    }
  }

  @Signal
  public void signal() {
    ready = true;
  }
}
```

RESULT	SAMPLES	SHARE	EXPECTED	MEANING
`STALE`	2	0.12%	Interesting	spins forever
`TERMINATED`	1,653	99.88%	Acceptable	saw it

With `volatile boolean ready`: `STALE` 0 of 3,308, declared forbidden.

- **Only 2 of 1,655 runs (0.12%) got stuck.** StaleFlag, which let the loop run for 1 s before setting the flag, got stuck in every run.
- **Lost updates**, from the guide’s own counter tests: `count++` on a plain field lost an update in 18.00% of 1,114,874,388 runs; `AtomicInteger` in 0 of 608,342,548.

- **Share mutable state through something that creates the edge:** `volatile`, an `Atomic*`, a lock, a concurrent collection, a future’s result, `Thread.start` / `join`.
- **Publish immutable objects through final fields** (JLS §17.5): a correctly constructed object’s final fields are seen initialised by every thread.
- **The bad outcomes were rare in jcstress** (0.12% stuck spins, 18.00% lost updates): a test that runs the code once will usually pass. jcstress runs it millions of times across compilation modes.

SOURCES JLS SE 21 §17.3, §17.4, §17.5 · jcstress 0.16: Mode.Termination, @Signal · results: deepdive/logs/staleflag.txt; ci-logs-ops logs/jcstress.log

10 · THE JAVA MEMORY MODEL

What volatile compiles to on x86-64

The memory model says what must be visible; the JIT decides how. For each volatile access C2 inserts memory-barrier nodes, and each CPU back end turns them into instructions — or into nothing, where the CPU already gives the ordering.

Where C2 puts the barriers — `barrierSetC2.cpp`

- **Volatile store**: `MemBarRelease` before it and `MemBarVolatile` after it, on CPUs that are multi-copy atomic (x86 defines `CPU_MULTI_COPY_ATOMIC`).
- **Volatile load**: `MemBarAcquire` after it.
- **Release store** (`VarHandle.setRelease`): `MemBarRelease` only.

An atomic add such as `getAndAdd` becomes `lock xadd` (`x86_64.ad`). `synchronized` runs with `LockingMode = 1` on the runner; `globals.hpp` documents 1 as "monitors & legacy stack-locking (LM_LEGACY, default)".

OpenJDK 21 hotspot/cpu/x86/x86_64.ad (excerpts)

```
instruct membar_acquire() %{
  match(MemBarAcquire);
  match(LoadFence);
  format %{ "MEMBAR-acquire ! (empty encoding)" %}

instruct membar_release() %{
  match(MemBarRelease);
  match(StoreFence);
  format %{ "MEMBAR-release ! (empty encoding)" %}

instruct membar_volatile(rFlagsReg cr) %{
  match(MemBarVolatile);
  format %{
    $$emit$$"lock addl [rsp + #0], 0\t! membar_volatile"
  %}
  ins_encode %{
    __ membar(Assembler::StoreLoad);
  %}
```

Measured

ONE THREAD, NO CONTENTION	NS/OP	ERROR
plain field: `plain++`	0.550	± 0.011
`setRelease` **(release store) +** `plain++`	0.740	± 0.004
volatile store + `plain++`	4.329	± 0.124
plain load	0.492	± 0.009
volatile load	0.504	± 0.011
`AtomicInteger.incrementAndGet`	2.377	± 0.026
`synchronized` **+** `++`**, uncontended**	6.200	± 0.021

FenceBench.java · JMH 1.37 · GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1

- **The volatile store** added 3.78 ns to the plain increment: the `lock addl` after it.
- **The release store** took 0.74 ns against 0.55 ns for the plain store, consistent with a barrier that emits no instruction on x86-64.
- **The volatile load** took 0.504 ns against 0.492 ns for a plain load: on x86-64 the acquire barrier is empty.
- **The atomic increment** (a `lock`-prefixed instruction) took 2.38 ns; an uncontended `synchronized` block 6.20 ns.

FenceBench.java

```
@Benchmark
public void plainStore() { plain++; }

@Benchmark
public void volatileStore() { vol = plain++; }

@Benchmark
public void releaseStore() {
  RELEASED.setRelease(this, plain++);
}

@Benchmark
public int plainLoad() { return plain; }

@Benchmark
public int volatileLoad() { return vol; }

@Benchmark
public int atomicIncrement() {
  return atomic.incrementAndGet();
}

@Benchmark
public void synchronizedIncrement() {
  synchronized (lock) {
    locked++;
  }
}
```

SOURCES OpenJDK 21 hotspot/share/gc/shared/c2/barrierSetC2.cpp, cpu/x86/x86_64.ad, cpu/x86/globalDefinitions_x86.hpp, share/utilities/globalDefinitions.hpp, runtime/globals.hpp (LockingMode) · results: deepdive/logs/jmh-fence.txt, jit-flags.txt

11 · KAFKA

Producer batching: linger.ms and batch.size

A Kafka producer does not send each record on its own. It appends records to a batch per partition and sends a batch when it is full or when it has waited `linger.ms`. The Kafka 4.3.1 client's documentation for `linger.ms` compares this to chapter 1:

`linger.ms`, from `ProducerConfig.java`

> "This can be thought of as analogous to Nagle's algorithm in TCP. … once we get `batch.size` worth of records for a partition it will be sent immediately regardless of this setting … The default changed from 0 to 5 in Apache Kafka 4.0 as the efficiency gains from larger batches typically result in similar or lower producer latency despite the increased linger."

SETTING	DEFAULT (4.3.1)	MEANING
`linger.ms`	5	how long a batch may wait for more records
`batch.size`	16384 bytes	a batch this full is sent at once
`acks`	all	the leader waits for the in-sync replicas
`enable.idempotence`	true	"exactly one copy of each message is written"
`fetch.min.bytes`	1	the broker answers a fetch as soon as 1 byte is available
`fetch.max.wait.ms`	500	or after this long, if less data is available
`max.poll.records`	500	most records returned by one `poll()`
`max.poll.interval.ms`	300000	longest gap between `poll()` calls

Defaults from the `define(…)` calls in `ProducerConfig.java` and `ConsumerConfig.java` at tag 4.3.1. `acks` is not measured here: with one broker the leader is the only in-sync replica, so `acks=all` and `acks=1` wait for the same write.

Measured: the same producer at two rates and three linger times

RATE	LINGER.MS	P50 MS	P99 MS	MAX MS	REQUESTS	RECORDS / REQUEST
20/s	0	2.33	6.92	10.99	106	1.0
20/s	5	7.01	8.97	9.90	105	1.0
20/s	50	8.30	53.13	58.30	55	2.0
2,000/s	0	1.71	32.82	50.51	7,273	1.4
2,000/s	5	3.06	6.95	12.24	915	11.0
2,000/s	50	25.71	50.22	51.53	105	99.0

KafkaLatency.java · 5 s per row, 100-byte records, one partition · latency from each record's due time to its acknowledgement · requests and records per request from the producer's own metrics (`request-total`, `records-per-request-avg`) · broker 4.3.1 in Docker · GitHub Actions · Linux 6.17.0-1022-azure · 4 CPUs · JDK 21.0.12.1

- **At 2,000 records/s**, `linger.ms=0` sent 7,273 requests (1.4 records each) and had a p99 of 32.8 ms. `linger.ms=5` sent 915 requests (11.0 records each) with a p99 of 7.0 ms — fewer requests and a lower tail, as the source's note on the 4.0 default change says.
- **At 2,000 records/s with `linger.ms=50`**: 99 records per request and a p50 of 25.7 ms — records waited for the batch.
- **At 20 records/s** (one every 50 ms) with `linger.ms=50`: p90 52.4 ms against 3.5 ms with `linger.ms=0`, and 2.0 records per request — a record with no others to batch waits up to the linger time, as the source says.

KafkaLatency.java — records sent on a schedule

```
for (int i = 0; i < n; i++) {
  long due = start + i * interval;
  while (System.nanoTime() < due) {
    Thread.onSpinWait();
  }
  final int k = i;
  prod.send(new ProducerRecord<>(t, new byte[100]),
      (md, e) -> {
        lat[k] = System.nanoTime() - due;
        done.countDown();
      });
}
```

SOURCES Apache Kafka 4.3.1: clients/.../producer/ProducerConfig.java (linger.ms, batch.size, acks, enable.idempotence) · results: deepdive/logs/kafka-producer.txt, kafka-broker.txt

11 · KAFKA

Consumer fetching, partitions and max.poll.interval.ms

`fetch.min.bytes` and `fetch.max.wait.ms`

> "If insufficient data is available the request will wait for that much data to accumulate before answering the request." `fetch.max.wait.ms` : "The maximum amount of time the server will block before answering the fetch request" when there is not enough data for `fetch.min.bytes` . — `ConsumerConfig.java`

Measured: one record every 100 ms

CONSUMER	P50 MS	P90 MS	P99 MS	MAX MS
fetch.min.bytes 1 (default)	3.0	3.7	52.3	52.3
fetch.min.bytes 1000000, max.wait 500	212.9	414.6	418.0	418.0

- **With the default `fetch.min.bytes=1`** the broker answered as soon as a record arrived: p50 3.0 ms from send to `poll()` .
- **With `fetch.min.bytes=1000000`** one record never filled a fetch, so the broker held each fetch until `fetch.max.wait.ms` : p50 213 ms, max 418 ms. A record that arrived early in a waiting fetch waited longest.

Measured: 4 consumers, 2 partitions

CONSUMER	PARTITIONS ASSIGNED
c0	1
c1	1
c2	0
c3	0

The `KafkaConsumer` Javadoc: the group divides the partitions "so that each partition is assigned to exactly one consumer in the group". Here 2 of the 4 consumers had nothing to read.

Measured: processing longer than `max.poll.interval.ms`

real output

```
max.poll.interval.ms 3000: polled 1 record, processing for 6 s
commitSync: CommitFailedException: Offset commit cannot be
completed since the consumer is not part of an active group for
auto partition assignment; it is likely that the consumer was
kicked out of the group.
```

The client documentation: "If poll() is not called before expiration of this timeout, then the consumer is considered failed and the group will rebalance in order to reassign the partitions to another member." The 6 s of processing exceeded the 3 s set here; the commit was then refused. The default is 300,000 ms.

KafkaLatency.java — slow processing

```
static void pollInterval() throws Exception {
  String t = topic(1);
  try (var prod = producer(0)) {
    prod.send(new ProducerRecord<>(t, new byte[100])).get();
  }
  try (var cons = consumer(Map.of("group.id", "slow-" + t,
      "auto.offset.reset", "earliest",
      "enable.auto.commit", false,
      "max.poll.interval.ms", 3000))) {
    cons.subscribe(Set.of(t));
    int got = 0;
    while (got == 0) {
      got = cons.poll(Duration.ofSeconds(1)).count();
    }
    System.out.println("max.poll.interval.ms 3000: polled "
        + got + " record, processing for 6 s");
    Thread.sleep(6_000);                    // "processing"
    try {
      cons.commitSync();
      System.out.println("commitSync: committed");
    } catch (Exception e) {
      System.out.println("commitSync: "
          + e.getClass().getSimpleName() + ": "
          + e.getMessage().lines().findFirst().orElse(""));
    }
  }
}
```

- **End-to-end latency includes the consumer's fetch wait.** Raising `fetch.min.bytes` trades latency for fewer fetch responses; the wait is bounded by `fetch.max.wait.ms` .
- **The number of partitions caps the number of consumers that do work** in one group.
- **Keep the time to process one `poll()` 's records below `max.poll.interval.ms`** , by processing faster or lowering `max.poll.records` (default 500); otherwise the consumer leaves the group and its commit fails.

SOURCES Apache Kafka 4.3.1: clients/.../consumer/ConsumerConfig.java, KafkaConsumer.java (Javadoc), CommonClientConfigs.java (MAX_POLL_INTERVAL_MS_DOC) · results: deepdive/logs/kafka-fetch.txt, kafka-partitions.txt, kafka-pollinterval.txt

BACK MATTER

Sources

Everything this book relies on for how things work. Measured results come from the logs listed last.

Standards and specifications

- RFC 9293 — Transmission Control Protocol (2022): Nagle (§3.7.4), delayed ACK (§3.8.6.3), TIME-WAIT and MSL (§3.4.2–3.4.3), states (§3.3.2)
- RFC 896 — J. Nagle, Congestion Control in IP/TCP Internetworks (1984)
- RFC 5737 — IPv4 address blocks reserved for documentation (192.0.2.0/24)
- JLS SE 21, chapter 17 — threads and locks; the memory model (§17.3–17.5)
- JEP 439 Generational ZGC · JEP 444 Virtual threads · JEP 474 ZGC generational by default · JEP 490 non-generational ZGC removed · JEP 491 synchronized without pinning

Linux 6.17 source and documentation

- include/net/tcp.h — TCP_DELACK_MIN/MAX, TCP_MAX_QUICKACKS, TCP_TIMEWAIT_LEN
- net/ipv4/tcp_input.c (quick ACKs, pingpong mode) · net/ipv4/tcp_output.c (tcp_event_data_sent) · net/ipv4/tcp.c (zero-linger close)
- Documentation/networking/ip-sysctl.rst — tcp_pingpong_thresh, tcp_tw_reuse, tcp_fin_timeout, tcp_syn_retries, ip_local_port_range
- Documentation/admin-guide/cgroup-v2.rst — cpu.max, cpu.stat, memory.max

OpenJDK 21 source

- java.net.http: PlainHttpConnection, MultiExchange, ResponseTimerEvent, module-info (keep-alive 30 s)
- java.base: StandardSocketOptions, Socket, URLConnection, VirtualThread, Object, FileInputStream, Blocker, ForkJoinPool, java.util.concurrent package-info, Continuation, NioSocketImpl, StackChunk
- jdk.httpserver ServerConfig · jdk.net ExtendedSocketOptions · jdk.jfr default.jfc
- HotSpot: compilationPolicy.hpp, compileTask.cpp, os.cpp, gcConfig.cpp, gc_globals.hpp, arguments.cpp, workerPolicy.cpp, cgroupUtil_linux.cpp, globals_linux.hpp, nmethod.cpp, zDriver.cpp, allocation.hpp, nmtDCmd.hpp, c2_globals.hpp, doCall.cpp, heapRegion.cpp, g1RemSet.hpp, g1BarrierSet.hpp, zAddress.hpp, zBarrier.hpp, barrierSetC2.cpp, x86_64.ad, globals_linux_x86.hpp

Documentation

- Java SE 21 HotSpot GC Tuning Guide — the G1 cycle, evacuation failure, full GC; "Available Collectors"
- HttpClient.Builder and HttpRequest.Builder Javadoc — connect and request timeouts
- HikariCP wiki, "About Pool Sizing" · J. D. C. Little, "A Proof for the Queuing Formula: L = λW", Operations Research 9(3), 1961

Apache Kafka 4.3.1 client source

- ProducerConfig, ConsumerConfig, KafkaConsumer, CommonClientConfigs — defaults and documentation of every setting the chapter uses

Libraries (source, default branch or named version)

- Tomcat SocketProperties · Jetty 12.0 ServerConnector · Netty 4.1 DefaultSocketChannelConfig, MultithreadEventLoopGroup
- pgjdbc PGProperty · HikariCP HikariConfig · Lettuce SocketOptions, RedisURI · Kafka Selector, ProducerConfig

Tools

- JMH 1.37 · jcstress 0.16 · HdrHistogram (recordValueWithExpectedInterval) · wrk2 · k6, open and closed models · Gatling, injection profiles

The evidence in this repository

- Experiments: `deepdive/src/main/java/deep/` and `concurrency/src/main/java/shop/` (jcstress) · runner script `deepdive/run.sh`
- Logs: branch `ci-logs-deepdive` (`logs/`) and `ci-logs-ops` (`logs/jcstress.log`), rewritten on every push to main

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